

Three-Tank Process Control - FAT Validation

TIA Portal V20 / S7-1500 / WinCC Unified

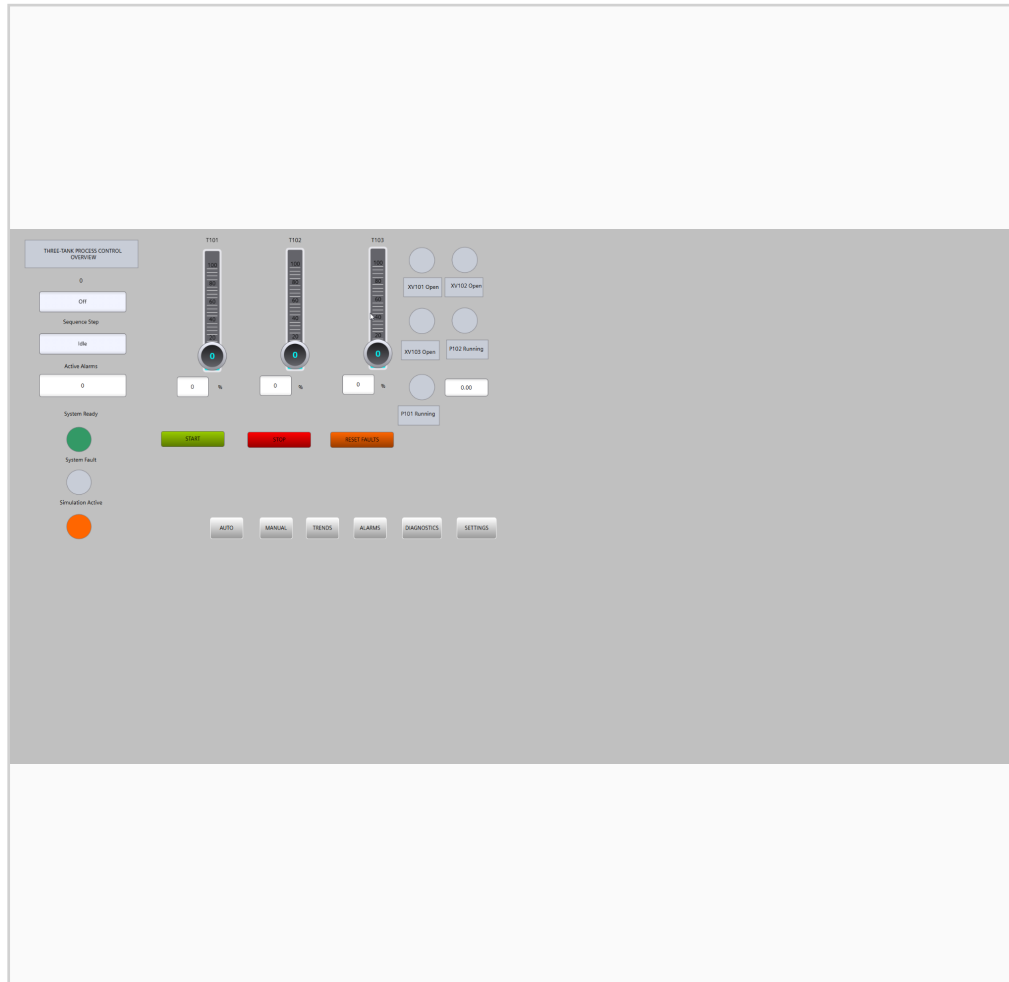
Format: requirement -> observed result -> retained screenshot evidence

Each test page pairs the checked behaviour with the observed result and retained HMI / PLC evidence.

Validation shown here was carried out with the project simulation and PLC/HMI evidence set.

Check	Safe deterministic startup
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Observed Step 0, ready state, tanks 0/0/0, no alarm/trip and no actuator request.



TT-F01-A

WinCC Overview in Off/Idle with T101/T102/T103 at 0, no active alarms, System Ready indicated.

Variable Name	Variable Type	Variable Value	Variable Unit	Variable Description
"DB_HM".Status.SystemReady	Bool	TRUE		
"DB_HM".Status.SystemControlEnabled	Bool	TRUE		
"DB_HM".Status.SystemActiveMode	DEC	2		0 OR, 1 Manual, 2 Auto
"DB_HM".Status.SystemAutoActive	Bool	TRUE		
"DB_HM".Status.SystemFaulted	Bool	FALSE		
"DB_HM".Status.SystemSimulationActive	Bool	TRUE		
"DB_HM".Status.SequenceActive	Bool	FALSE		
"DB_HM".Status.SequenceStep	DEC	10		
"DB_HM".Status.SequenceComplete	Bool	FALSE		
"DB_HM".Status.Alarms.AnyAlarm	Bool	FALSE		
"DB_HM".Status.Alarms.AnyTrip	Bool	FALSE		
"DB_HM".Status.Alarms.FirstOutCode	DEC	0		
"DB_IO".SelectedIn.TankLevelPct[1]	Floating-point num.	0.0		
"DB_IO".SelectedIn.TankLevelPct[2]	Floating-point num.	0.0		
"DB_IO".SelectedIn.TankLevelPct[3]	Floating-point num.	0.0		
"DB_IO".SelectedIn.Valve[1].OpenFb	Bool	FALSE		
"DB_IO".SelectedIn.Valve[1].ClosedFb	Bool	TRUE		
"DB_IO".SelectedIn.Valve[2].OpenFb	Bool	FALSE		
"DB_IO".SelectedIn.Valve[2].ClosedFb	Bool	TRUE		
"DB_IO".SelectedIn.Valve[3].OpenFb	Bool	FALSE		
"DB_IO".SelectedIn.Valve[3].ClosedFb	Bool	TRUE		
"DB_IO".SelectedIn.Pump[1].RunFb	Bool	FALSE		
"DB_IO".SelectedIn.Pump[1].OverloadOK	Bool	TRUE		
"DB_IO".SelectedIn.Pump[2].RunFb	Bool	FALSE		
"DB_IO".SelectedIn.Pump[2].OverloadOK	Bool	TRUE		
"DB_IO".ProcessOut.ValveOpenCmd[1]	Bool	FALSE		
"DB_IO".ProcessOut.ValveOpenCmd[2]	Bool	FALSE		
"DB_IO".ProcessOut.ValveOpenCmd[3]	Bool	FALSE		
"DB_IO".ProcessOut.PumpStartCmd[1]	Bool	FALSE		
"DB_IO".ProcessOut.PumpStartCmd[2]	Bool	FALSE		
"DB_HM".Status.Permissives.System	Bool	TRUE		
"DB_HM".Status.Permissives.AutoStart	Bool	FALSE		
"DB_HM".Status.Permissives.Valve[1]	Bool	TRUE		
"DB_HM".Status.Permissives.Valve[2]	Bool	FALSE		
"DB_HM".Status.Permissives.Valve[3]	Bool	FALSE		
"DB_HM".Status.Permissives.Pump[1]	Bool	FALSE		
"DB_HM".Status.Permissives.Pump[2]	Bool	FALSE		

TT-F01-B

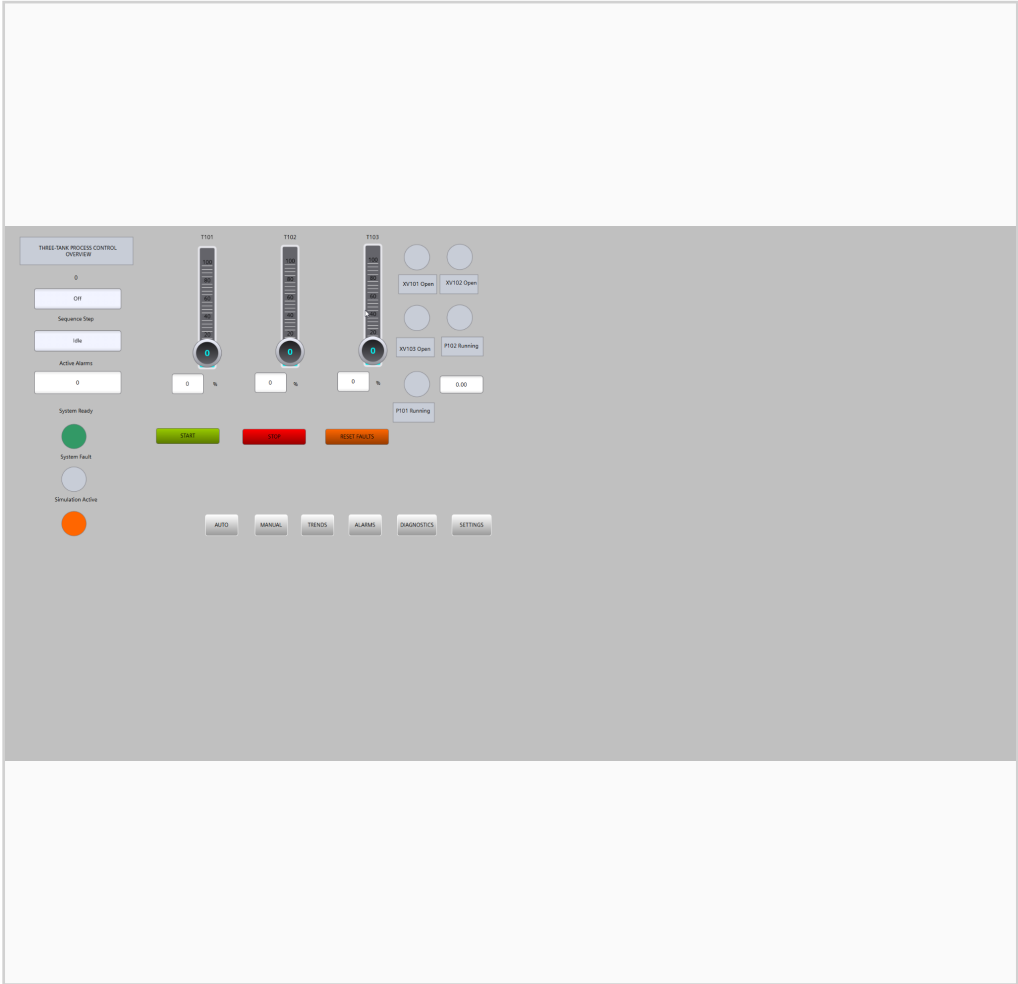
Watch table baseline: Step 0, no fault/trip, tank levels 0/0/0, process commands off.

- Check

Reject start outside a valid automatic context
- Observed

Controller stayed in Step 0 with commands off and no trip.

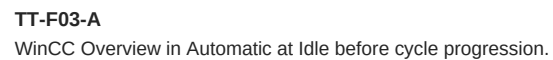
*DB_HMI".Status.System_Ready	Bool	☒	TRUE		
*DB_HMI".Status.System_ControlEnabled	Bool	☒	TRUE		
*DB_HMI".Status.System_ActiveMode	DEC		2		0 Off, 1 Manual, 2 Auto
*DB_HMI".Status.System_Active	Bool	☒	TRUE		
*DB_HMI".Status.System_Faulted	Bool	☐	FALSE		
*DB_HMI".Status.System_SimulationActive	Bool	☒	TRUE		
*DB_HMI".Status.Sequence_Active	Bool	☐	FALSE		
*DB_HMI".Status.Sequence_Step	DEC		10		
*DB_HMI".Status.Sequence_Complete	Bool	☐	FALSE		
*DB_HMI".Status.Alarms_AnyAlarm	Bool	☐	FALSE		
*DB_HMI".Status.Alarms_FirstOutCode	DEC		0		
*DB_IO".SelectedIn.TankLevelPct[1]	Floating point nu...		0.0		
*DB_IO".SelectedIn.TankLevelPct[2]	Floating point nu...		0.0		
*DB_IO".SelectedIn.TankLevelPct[3]	Floating point nu...		0.0		
*DB_IO".SelectedIn.Valve[1].OpenFb	Bool	☐	FALSE		
*DB_IO".SelectedIn.Valve[1].ClosedFb	Bool	☒	TRUE		
*DB_IO".SelectedIn.Valve[2].OpenFb	Bool	☐	FALSE		
*DB_IO".SelectedIn.Valve[2].ClosedFb	Bool	☒	TRUE		
*DB_IO".SelectedIn.Valve[3].OpenFb	Bool	☐	FALSE		
*DB_IO".SelectedIn.Valve[3].ClosedFb	Bool	☒	TRUE		
*DB_IO".SelectedIn.Pump[1].RunFb	Bool	☐	FALSE		
*DB_IO".SelectedIn.Pump[1].OverloadOK	Bool	☒	TRUE		
*DB_IO".SelectedIn.Pump[2].RunFb	Bool	☐	FALSE		
*DB_IO".SelectedIn.Pump[2].OverloadOK	Bool	☒	TRUE		
*DB_IO".ProcessOut.ValveOpenCmd[1]	Bool	☐	FALSE		
*DB_IO".ProcessOut.ValveOpenCmd[2]	Bool	☐	FALSE		
*DB_IO".ProcessOut.ValveOpenCmd[3]	Bool	☐	FALSE		
*DB_IO".ProcessOut.PumpStartCmd[1]	Bool	☐	FALSE		
*DB_IO".ProcessOut.PumpStartCmd[2]	Bool	☐	FALSE		
*DB_HMI".Status.Permissives_System	Bool	☒	TRUE		
*DB_HMI".Status.Permissives_AutoStart	Bool	☐	FALSE		
*DB_HMI".Status.Permissives_Valve[1]	Bool	☒	TRUE		
*DB_HMI".Status.Permissives_Valve[2]	Bool	☐	FALSE		
*DB_HMI".Status.Permissives_Valve[3]	Bool	☐	FALSE		
*DB_HMI".Status.Permissives_Pump[1]	Bool	☐	FALSE		
*DB_HMI".Status.Permissives_Pump[2]	Bool	☐	FALSE		
<add new>					



TT-F02-A
Step 0 / commands off / no trip baseline.

TT-F02-B
HMI Off/Idle with no sequence activity.

Observed Sequence progressed through precheck to S20; T101 began filling through XV101.



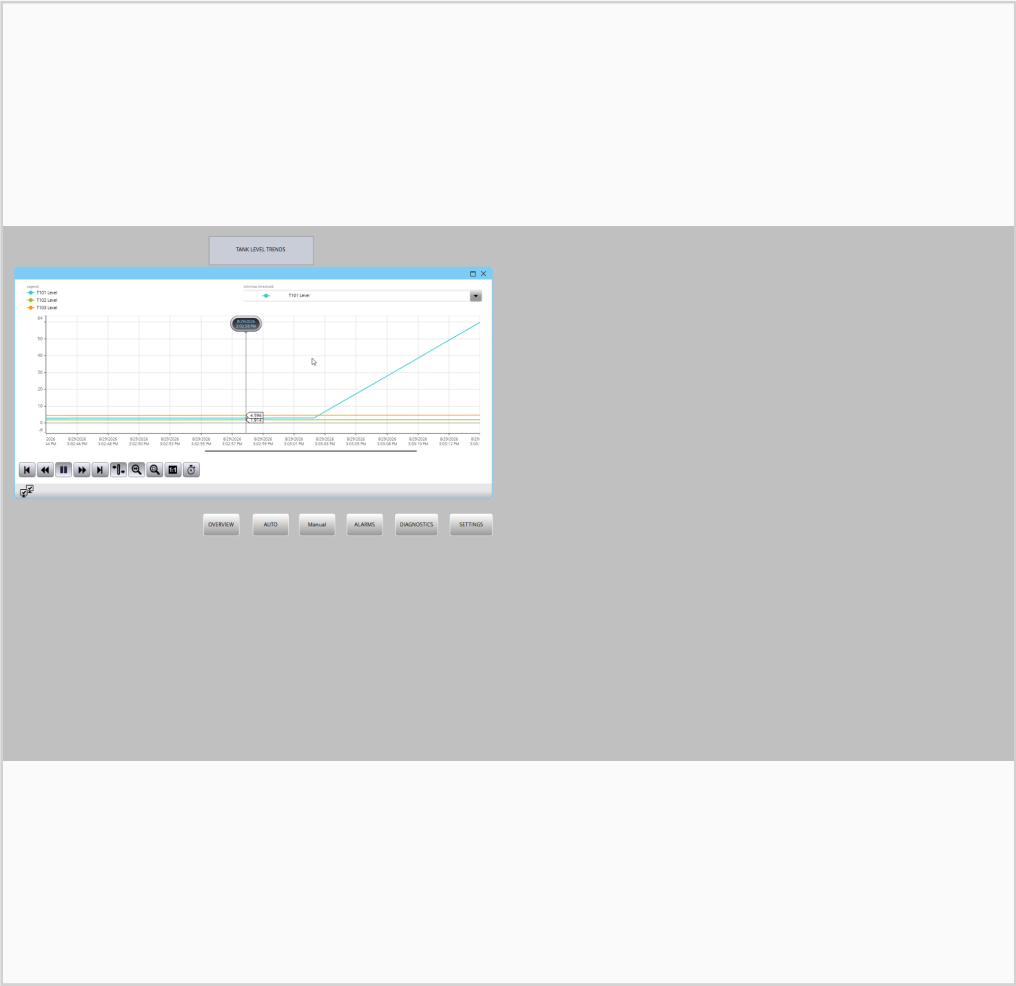
TT-F03-B
Watch table at Step 20 with T101 rising and XV101 process command active; no trip.

Check Verify S20 filling behaviour

Observed T101 rose toward its target with XV101 active and pumps not demanded.

*DB_HMI".Status.System.Ready	Bool	☒	TRUE		
*DB_HMI".Status.System.ControlEnabled	Bool	☒	TRUE		
*DB_HMI".Status.System.ActiveMode	DEC		2		0 Off, 1 Manual, 2 Auto
*DB_HMI".Status.System.AutoActive	Bool	☒	TRUE		
*DB_HMI".Status.System.Faulted	Bool	☐	FALSE		
*DB_HMI".Status.System.SimulationActive	Bool	☒	TRUE		
*DB_HMI".Status.Sequence.Active	Bool	☐	FALSE		
*DB_HMI".Status.Sequence.Step	DEC		20		
*DB_HMI".Status.Sequence.Complete	Bool	☐	FALSE		
*DB_HMI".Status.Alarms.AnyAlarm	Bool	☐	FALSE		
*DB_HMI".Status.Alarms.FirstOutCode	DEC		0		
*DB_IO".SelectedIn.TankLevelPct[1]	Floating point nu...		0.0		
*DB_IO".SelectedIn.TankLevelPct[2]	Floating point nu...		0.0		
*DB_IO".SelectedIn.TankLevelPct[3]	Floating point nu...		0.0		
*DB_IO".SelectedIn.Valve[1].OpenFb	Bool	☒	TRUE		
*DB_IO".SelectedIn.Valve[1].ClosedFb	Bool	☐	FALSE		
*DB_IO".SelectedIn.Valve[2].OpenFb	Bool	☐	FALSE		
*DB_IO".SelectedIn.Valve[2].ClosedFb	Bool	☒	TRUE		
*DB_IO".SelectedIn.Valve[3].OpenFb	Bool	☐	FALSE		
*DB_IO".SelectedIn.Valve[3].ClosedFb	Bool	☒	TRUE		
*DB_IO".SelectedIn.Pump[1].RunFb	Bool	☐	FALSE		
*DB_IO".SelectedIn.Pump[1].OverloadOK	Bool	☒	TRUE		
*DB_IO".SelectedIn.Pump[2].RunFb	Bool	☐	FALSE		
*DB_IO".SelectedIn.Pump[2].OverloadOK	Bool	☒	TRUE		
*DB_IO".ProcessOut.ValveOpenCmd[1]	Bool	☒	TRUE		
*DB_IO".ProcessOut.ValveOpenCmd[2]	Bool	☐	FALSE		
*DB_IO".ProcessOut.ValveOpenCmd[3]	Bool	☐	FALSE		
*DB_IO".ProcessOut.PumpStartCmd[1]	Bool	☐	FALSE		
*DB_IO".ProcessOut.PumpStartCmd[2]	Bool	☐	FALSE		
*DB_HMI".Status.Permissives.System	Bool	☒	TRUE		
*DB_HMI".Status.Permissives.AutoStart	Bool	☐	FALSE		
*DB_HMI".Status.Permissives.Valve[1]	Bool	☒	TRUE		
*DB_HMI".Status.Permissives.Valve[2]	Bool	☐	FALSE		
*DB_HMI".Status.Permissives.Valve[3]	Bool	☐	FALSE		
*DB_HMI".Status.Permissives.Pump[1]	Bool	☐	FALSE		
*DB_HMI".Status.Permissives.Pump[2]	Bool	☐	FALSE		
-add new-					

TT-F04-A
Step 20, T101 increasing, XV101 command/feedback active, pumps off.



TT-F04-B
WinCC tank-level trend showing the T101 fill ramp.

Check Prove XV102 before pump demand

Observed At S40 entry XV102 command was active while OpenFb was still false; P101 remained off.

"DB_HMI".Status.System.Ready	Bool	☒	TRUE						
"DB_HMI".Status.System.ControlEnabled	Bool	☒	TRUE						
"DB_HMI".Status.System.ActiveMode	DEC		2					0 Off, 1 Manual, 2 Auto	
"DB_HMI".Status.System.AutoActive	Bool	☒	TRUE						
"DB_HMI".Status.System.Faulted	Bool	☐	FALSE						
"DB_HMI".Status.System.SimulationActive	Bool	☒	TRUE						
"DB_HMI".Status.Sequence.Active	Bool	☐	FALSE						
"DB_HMI".Status.Sequence.Step	DEC+/-		40						
"DB_HMI".Status.Sequence.Complete	Bool	☐	FALSE						
"DB_HMI".Status.Alarms.AnyAlarm	Bool	☐	FALSE						
"DB_HMI".Status.Alarms.AnyTrip	Bool	☐	FALSE						
"DB_HMI".Status.Alarms.FirstOutCode	DEC		0						
"DB_IO".SelectedIn.TankLevelPct[1]	Floating-point nu...		80.0						
"DB_IO".SelectedIn.TankLevelPct[2]	Floating-point nu...		0.0						
"DB_IO".SelectedIn.TankLevelPct[3]	Floating-point nu...		0.0						
"DB_IO".SelectedIn.Valve[1].OpenFb	Bool	☐	FALSE						
"DB_IO".SelectedIn.Valve[1].ClosedFb	Bool	☒	TRUE						
"DB_IO".SelectedIn.Valve[2].OpenFb	Bool	☐	FALSE						
"DB_IO".SelectedIn.Valve[2].ClosedFb	Bool	☐	FALSE						
"DB_IO".SelectedIn.Valve[3].OpenFb	Bool	☐	FALSE						
"DB_IO".SelectedIn.Valve[3].ClosedFb	Bool	☒	TRUE						
"DB_IO".SelectedIn.Pump[1].RunFb	Bool	☐	FALSE						
"DB_IO".SelectedIn.Pump[1].OverloadOK	Bool	☒	TRUE						
"DB_IO".SelectedIn.Pump[2].RunFb	Bool	☐	FALSE						
"DB_IO".SelectedIn.Pump[2].OverloadOK	Bool	☒	TRUE						
"DB_IO".ProcessOut.ValveOpenCmd[1]	Bool	☐	FALSE						
"DB_IO".ProcessOut.ValveOpenCmd[2]	Bool	☒	TRUE						
"DB_IO".ProcessOut.ValveOpenCmd[3]	Bool	☐	FALSE						
"DB_IO".ProcessOut.PumpStartCmd[1]	Bool	☐	FALSE						
"DB_IO".ProcessOut.PumpStartCmd[2]	Bool	☐	FALSE						
"DB_HMI".Status.Permissives.System	Bool	☒	TRUE						
"DB_HMI".Status.Permissives.AutoStart	Bool	☐	FALSE						
"DB_HMI".Status.Permissives.Valve[1]	Bool	☒	TRUE						
"DB_HMI".Status.Permissives.Valve[2]	Bool	☒	TRUE						
"DB_HMI".Status.Permissives.Valve[3]	Bool	☐	FALSE						
"DB_HMI".Status.Permissives.Pump[1]	Bool	☐	FALSE						
"DB_HMI".Status.Permissives.Pump[2]	Bool	☐	FALSE						
<Add new>									

TT-F05-A

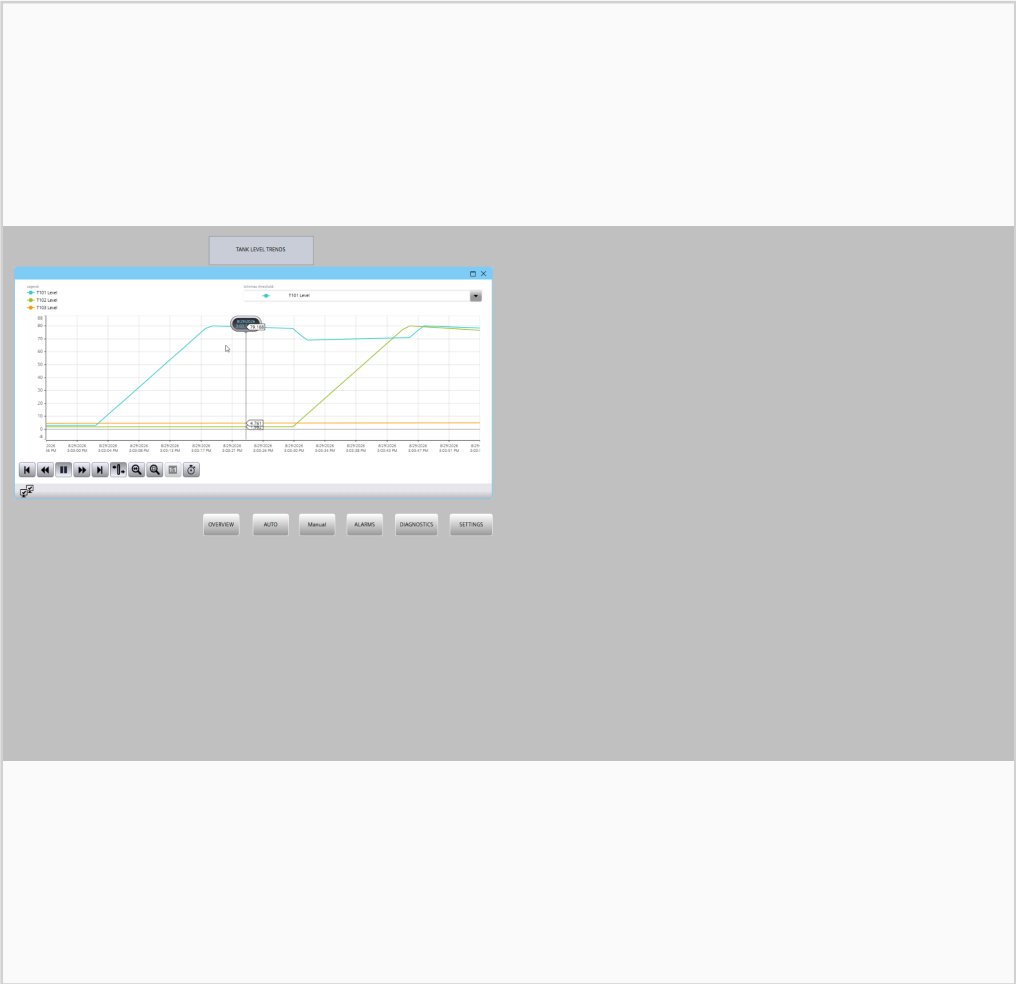
Step 40 entry: ValveOpenCmd[2]=TRUE while XV102 OpenFb=FALSE and PumpStartCmd[1]=FALSE.

Check Verify T101 -> T102 transfer behaviour

Observed T102 rose during transfer while upstream level behaviour followed the process model.

*DB_HMI4.Status.SystemReady	Bool	☒	TRUE		
*DB_HMI4.Status.SystemControlEnabled	Bool	☒	TRUE		
*DB_HMI4.Status.SystemActiveMode	DEC		2		0 Off, 1 Manual, 2 Auto
*DB_HMI4.Status.SystemAutoActive	Bool	☒	TRUE		
*DB_HMI4.Status.SystemFaulted	Bool	☐	FALSE		
*DB_HMI4.Status.SystemSimulationActive	Bool	☒	TRUE		
*DB_HMI4.Status.SequenceActive	Bool	☐	FALSE		
*DB_HMI4.Status.SequenceStep	DEC		40		
*DB_HMI4.Status.SequenceComplete	Bool	☐	FALSE		
*DB_HMI4.Status.AlarmsAnyAlarm	Bool	☐	FALSE		
*DB_HMI4.Status.AlarmsFirstOutCode	DEC		0		
*DB_IO4.SelectedIn.TankLevelPct[1]	Floating point nu...		74.5		
*DB_IO4.SelectedIn.TankLevelPct[2]	Floating point nu...		80.0		
*DB_IO4.SelectedIn.TankLevelPct[3]	Floating point nu...		0.0		
*DB_IO4.SelectedIn.Valve[1].OpenFb	Bool	☒	TRUE		
*DB_IO4.SelectedIn.Valve[1].ClosedFb	Bool	☐	FALSE		
*DB_IO4.SelectedIn.Valve[2].OpenFb	Bool	☐	FALSE		
*DB_IO4.SelectedIn.Valve[2].ClosedFb	Bool	☐	FALSE		
*DB_IO4.SelectedIn.Valve[3].OpenFb	Bool	☐	FALSE		
*DB_IO4.SelectedIn.Valve[3].ClosedFb	Bool	☒	TRUE		
*DB_IO4.SelectedIn.Pump[1].RunFb	Bool	☒	TRUE		
*DB_IO4.SelectedIn.Pump[1].OverloadOK	Bool	☒	TRUE		
*DB_IO4.SelectedIn.Pump[2].RunFb	Bool	☐	FALSE		
*DB_IO4.SelectedIn.Pump[2].OverloadOK	Bool	☒	TRUE		
*DB_IO4.ProcessOut.ValveOpenCmd[1]	Bool	☒	TRUE		
*DB_IO4.ProcessOut.ValveOpenCmd[2]	Bool	☐	FALSE		
*DB_IO4.ProcessOut.ValveOpenCmd[3]	Bool	☐	FALSE		
*DB_IO4.ProcessOut.PumpStartCmd[1]	Bool	☐	FALSE		
*DB_IO4.ProcessOut.PumpStartCmd[2]	Bool	☒	TRUE		
*DB_HMI4.Status.Permissives.System	Bool	☒	TRUE		
*DB_HMI4.Status.Permissives.AutoStart	Bool	☐	FALSE		
*DB_HMI4.Status.Permissives.Valve[1]	Bool	☒	TRUE		
*DB_HMI4.Status.Permissives.Valve[2]	Bool	☒	TRUE		
*DB_HMI4.Status.Permissives.Valve[3]	Bool	☒	TRUE		
*DB_HMI4.Status.Permissives.Pump[1]	Bool	☐	FALSE		
*DB_HMI4.Status.Permissives.Pump[2]	Bool	☐	FALSE		
+add new+					

TT-F06-A
Step 40 transfer-state capture with changing T101/T102 levels and healthy permissives.



TT-F06-B
WinCC trend showing T101/T102 transfer behavior across the cycle.

Check Finish transfer cleanly

Observed At S50 T101/T102 were at target and transfer commands were removed.

"DB_HMI".Status.System.Ready	Bool	☒	TRUE		☐			
"DB_HMI".Status.System.ControlEnabled	Bool	☒	TRUE		☐			
"DB_HMI".Status.System.ActiveMode	DEC		2		☐		0 Off, 1 Manual, 2 Auto	
"DB_HMI".Status.System.AutoActive	Bool	☒	TRUE		☐			
"DB_HMI".Status.System.Faulted	Bool	☐	FALSE		☐			
"DB_HMI".Status.System.SimulationActive	Bool	☒	TRUE		☐			
"DB_HMI".Status.Sequence.Active	Bool	☐	FALSE		☐			
"DB_HMI".Status.Sequence.Step	DEC+/-		50		☐			
"DB_HMI".Status.Sequence.Complete	Bool	☐	FALSE		☐			
"DB_HMI".Status.Alarms.AnyAlarm	Bool	☐	FALSE		☐			
"DB_HMI".Status.Alarms.AnyTrip	Bool	☐	FALSE		☐			
"DB_HMI".Status.Alarms.FirstOutCode	DEC		0		☐			
"DB_IO".SelectedIn.TankLevelPct[1]	Floating-point nu...		80.0		☐			
"DB_IO".SelectedIn.TankLevelPct[2]	Floating-point nu...		80.0		☐			
"DB_IO".SelectedIn.TankLevelPct[3]	Floating-point nu...		0.0		☐			
"DB_IO".SelectedIn.Valve[1].OpenFb	Bool	☐	FALSE		☐			
"DB_IO".SelectedIn.Valve[1].ClosedFb	Bool	☒	TRUE		☐			
"DB_IO".SelectedIn.Valve[2].OpenFb	Bool	☐	FALSE		☐			
"DB_IO".SelectedIn.Valve[2].ClosedFb	Bool	☒	TRUE		☐			
"DB_IO".SelectedIn.Valve[3].OpenFb	Bool	☐	FALSE		☐			
"DB_IO".SelectedIn.Valve[3].ClosedFb	Bool	☒	TRUE		☐			
"DB_IO".SelectedIn.Pump[1].RunFb	Bool	☐	FALSE		☐			
"DB_IO".SelectedIn.Pump[1].OverloadOK	Bool	☒	TRUE		☐			
"DB_IO".SelectedIn.Pump[2].RunFb	Bool	☐	FALSE		☐			
"DB_IO".SelectedIn.Pump[2].OverloadOK	Bool	☒	TRUE		☐			
"DB_IO".ProcessOut.ValveOpenCmd[1]	Bool	☐	FALSE		☐			
"DB_IO".ProcessOut.ValveOpenCmd[2]	Bool	☐	FALSE		☐			
"DB_IO".ProcessOut.ValveOpenCmd[3]	Bool	☐	FALSE		☐			
"DB_IO".ProcessOut.PumpStartCmd[1]	Bool	☐	FALSE		☐			
"DB_IO".ProcessOut.PumpStartCmd[2]	Bool	☐	FALSE		☐			
"DB_HMI".Status.Permissives.System	Bool	☒	TRUE		☐			
"DB_HMI".Status.Permissives.AutoStart	Bool	☐	FALSE		☐			
"DB_HMI".Status.Permissives.Valve[1]	Bool	☒	TRUE		☐			
"DB_HMI".Status.Permissives.Valve[2]	Bool	☒	TRUE		☐			
"DB_HMI".Status.Permissives.Valve[3]	Bool	☒	TRUE		☐			
"DB_HMI".Status.Permissives.Pump[1]	Bool	☐	FALSE		☐			
"DB_HMI".Status.Permissives.Pump[2]	Bool	☐	FALSE		☐			
<Add new>					☐			

TT-F07-A

Step 50 with T101=80 and T102=80; process valve/pump commands off.

Check Require XV103 path before P102 operation

Observed During the live test P102 remained inhibited until the XV103 path was proved. The retained S60 frame shows the resulting valid run state with XV103 feedback established.

"DB_HMI".Status.System.Ready	Bool	☒	TRUE		☐			
"DB_HMI".Status.System.ControlEnabled	Bool	☒	TRUE		☐			
"DB_HMI".Status.System.ActiveMode	DEC		2		☐		0 Off, 1 Manual, 2 Auto	
"DB_HMI".Status.System.AutoActive	Bool	☒	TRUE		☐			
"DB_HMI".Status.System.Faulted	Bool	☐	FALSE		☐			
"DB_HMI".Status.System.SimulationActive	Bool	☒	TRUE		☐			
"DB_HMI".Status.Sequence.Active	Bool	☐	FALSE		☐			
"DB_HMI".Status.Sequence.Step	DEC+/-		60		☐			
"DB_HMI".Status.Sequence.Complete	Bool	☐	FALSE		☐			
"DB_HMI".Status.Alarms.AnyAlarm	Bool	☐	FALSE		☐			
"DB_HMI".Status.Alarms.AnyTrip	Bool	☐	FALSE		☐			
"DB_HMI".Status.Alarms.FirstOutCode	DEC		0		☐			
"DB_IO".SelectedIn.TankLevelPct[1]	Floating-point nu...		69.0		☐			
"DB_IO".SelectedIn.TankLevelPct[2]	Floating-point nu...		58.5		☐			
"DB_IO".SelectedIn.TankLevelPct[3]	Floating-point nu...		40.0		☐			
"DB_IO".SelectedIn.Valve[1].OpenFb	Bool	☒	TRUE		☐			
"DB_IO".SelectedIn.Valve[1].ClosedFb	Bool	☐	FALSE		☐			
"DB_IO".SelectedIn.Valve[2].OpenFb	Bool	☒	TRUE		☐			
"DB_IO".SelectedIn.Valve[2].ClosedFb	Bool	☐	FALSE		☐			
"DB_IO".SelectedIn.Valve[3].OpenFb	Bool	☒	TRUE		☐			
"DB_IO".SelectedIn.Valve[3].ClosedFb	Bool	☐	FALSE		☐			
"DB_IO".SelectedIn.Pump[1].RunFb	Bool	☒	TRUE		☐			
"DB_IO".SelectedIn.Pump[1].OverloadOK	Bool	☒	TRUE		☐			
"DB_IO".SelectedIn.Pump[2].RunFb	Bool	☒	TRUE		☐			
"DB_IO".SelectedIn.Pump[2].OverloadOK	Bool	☒	TRUE		☐			
"DB_IO".ProcessOut.ValveOpenCmd[1]	Bool	☒	TRUE		☐			
"DB_IO".ProcessOut.ValveOpenCmd[2]	Bool	☒	TRUE		☐			
"DB_IO".ProcessOut.ValveOpenCmd[3]	Bool	☒	TRUE		☐			
"DB_IO".ProcessOut.PumpStartCmd[1]	Bool	☒	TRUE		☐			
"DB_IO".ProcessOut.PumpStartCmd[2]	Bool	☒	TRUE		☐			
"DB_HMI".Status.Permissives.System	Bool	☒	TRUE		☐			
"DB_HMI".Status.Permissives.AutoStart	Bool	☐	FALSE		☐			
"DB_HMI".Status.Permissives.Valve[1]	Bool	☒	TRUE		☐			
"DB_HMI".Status.Permissives.Valve[2]	Bool	☒	TRUE		☐			
"DB_HMI".Status.Permissives.Valve[3]	Bool	☒	TRUE		☐			
"DB_HMI".Status.Permissives.Pump[1]	Bool	☒	TRUE		☐			
"DB_HMI".Status.Permissives.Pump[2]	Bool	☒	TRUE		☐			
<Add new>					☐			

TT-F08-A

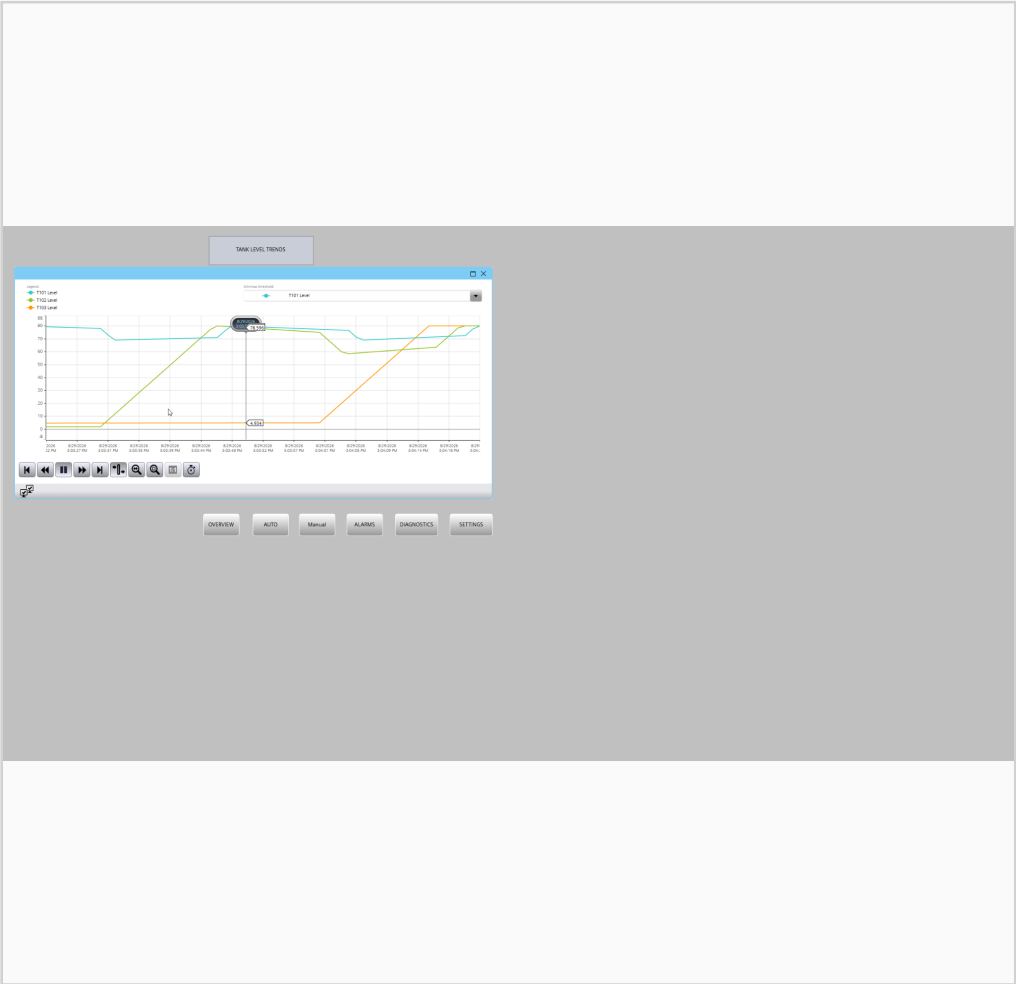
Step 60 running with XV103 open feedback and P102 RunFb/command active.

Check Verify cascade fill to T103

Observed T103 rose while the upstream tanks were replenished/transferred according to the sequence.

*DB_HMI".Status.SystemReady	Bool	☒	TRUE		
*DB_HMI".Status.SystemControlEnabled	Bool	☒	TRUE		
*DB_HMI".Status.System.ActiveMode	DEC		2		0 Off, 1 Manual, 2 Auto
*DB_HMI".Status.System.AutoActive	Bool	☒	TRUE		
*DB_HMI".Status.System.Faulted	Bool	☐	FALSE		
*DB_HMI".Status.System.SimulationActive	Bool	☒	TRUE		
*DB_HMI".Status.Sequence.Active	Bool	☐	FALSE		
*DB_HMI".Status.Sequence.Step	DEC		60		
*DB_HMI".Status.Sequence.Complete	Bool	☐	FALSE		
*DB_HMI".Status.Alarms.AnyAlarm	Bool	☐	FALSE		
*DB_HMI".Status.Alarms.AnyTrip	Bool	☐	FALSE		
*DB_HMI".Status.Alarms.FirstOutCode	DEC		0		
*DB_IO".SelectedIn.TankLevelPct[1]	Floating point nu...		69.0		
*DB_IO".SelectedIn.TankLevelPct[2]	Floating point nu...		58.5		
*DB_IO".SelectedIn.TankLevelPct[3]	Floating point nu...		40.0		
*DB_IO".SelectedIn.Valve[1].OpenFb	Bool	☒	TRUE		
*DB_IO".SelectedIn.Valve[1].ClosedFb	Bool	☐	FALSE		
*DB_IO".SelectedIn.Valve[2].OpenFb	Bool	☒	TRUE		
*DB_IO".SelectedIn.Valve[2].ClosedFb	Bool	☐	FALSE		
*DB_IO".SelectedIn.Valve[3].OpenFb	Bool	☒	TRUE		
*DB_IO".SelectedIn.Valve[3].ClosedFb	Bool	☐	FALSE		
*DB_IO".SelectedIn.Pump[1].RunFb	Bool	☒	TRUE		
*DB_IO".SelectedIn.Pump[1].OverloadOK	Bool	☒	TRUE		
*DB_IO".SelectedIn.Pump[2].RunFb	Bool	☒	TRUE		
*DB_IO".SelectedIn.Pump[2].OverloadOK	Bool	☒	TRUE		
*DB_IO".ProcessOut.ValveOpenCmd[1]	Bool	☒	TRUE		
*DB_IO".ProcessOut.ValveOpenCmd[2]	Bool	☒	TRUE		
*DB_IO".ProcessOut.ValveOpenCmd[3]	Bool	☒	TRUE		
*DB_IO".ProcessOut.PumpStartCmd[1]	Bool	☒	TRUE		
*DB_IO".ProcessOut.PumpStartCmd[2]	Bool	☒	TRUE		
*DB_HMI".Status.Permissives.System	Bool	☒	TRUE		
*DB_HMI".Status.Permissives.AutoStart	Bool	☐	FALSE		
*DB_HMI".Status.Permissives.Valve[1]	Bool	☒	TRUE		
*DB_HMI".Status.Permissives.Valve[2]	Bool	☒	TRUE		
*DB_HMI".Status.Permissives.Valve[3]	Bool	☒	TRUE		
*DB_HMI".Status.Permissives.Pump[1]	Bool	☒	TRUE		
*DB_HMI".Status.Permissives.Pump[2]	Bool	☒	TRUE		
+add new-					

TT-F09-A
Step 60 with T103 rising and upstream refill/transfer commands active.



TT-F09-B
WinCC full-cycle trend showing T103 rise and upstream level recovery/maintenance.

- Check
- Complete the automatic cycle
- Observed
- S70 captured with all tanks at target, process commands off, Complete true and Active false.

*DB_HMI".Status.System.Ready	Bool	☒	TRUE		
*DB_HMI".Status.System.ControlEnabled	Bool	☒	TRUE		
*DB_HMI".Status.System.ActiveMode	DEC		2		0 Off, 1 Manual, 2 Auto
*DB_HMI".Status.System.AutoActive	Bool	☒	TRUE		
*DB_HMI".Status.System.Faulted	Bool	☐	FALSE		
*DB_HMI".Status.System.SimulationActive	Bool	☒	TRUE		
*DB_HMI".Status.Sequence.Active	Bool	☐	FALSE		
*DB_HMI".Status.Sequence.Step	DEC+		70		
*DB_HMI".Status.Sequence.Complete	Bool	☐	FALSE		
*DB_HMI".Status.Alarms.AnyAlarm	Bool	☐	FALSE		
*DB_HMI".Status.Alarms.AnyTrip	Bool	☐	FALSE		
*DB_HMI".Status.Alarms.FirstOutCode	DEC		0		
*DB_IO".SelectedIn.TankLevelPct[1]	Floating point nu...		80.0		
*DB_IO".SelectedIn.TankLevelPct[2]	Floating point nu...		80.0		
*DB_IO".SelectedIn.TankLevelPct[3]	Floating point nu...		80.0		
*DB_IO".SelectedIn.Valve[1].OpenFb	Bool	☐	FALSE		
*DB_IO".SelectedIn.Valve[1].ClosedFb	Bool	☒	TRUE		
*DB_IO".SelectedIn.Valve[2].OpenFb	Bool	☐	FALSE		
*DB_IO".SelectedIn.Valve[2].ClosedFb	Bool	☒	TRUE		
*DB_IO".SelectedIn.Valve[3].OpenFb	Bool	☐	FALSE		
*DB_IO".SelectedIn.Valve[3].ClosedFb	Bool	☒	TRUE		
*DB_IO".SelectedIn.Pump[1].RunFb	Bool	☐	FALSE		
*DB_IO".SelectedIn.Pump[1].OverloadOK	Bool	☒	TRUE		
*DB_IO".SelectedIn.Pump[2].RunFb	Bool	☐	FALSE		
*DB_IO".SelectedIn.Pump[2].OverloadOK	Bool	☒	TRUE		
*DB_IO".ProcessOut.ValveOpenCmd[1]	Bool	☐	FALSE		
*DB_IO".ProcessOut.ValveOpenCmd[2]	Bool	☐	FALSE		
*DB_IO".ProcessOut.ValveOpenCmd[3]	Bool	☐	FALSE		
*DB_IO".ProcessOut.PumpStartCmd[1]	Bool	☐	FALSE		
*DB_IO".ProcessOut.PumpStartCmd[2]	Bool	☐	FALSE		
*DB_HMI".Status.Permissives.System	Bool	☒	TRUE		
*DB_HMI".Status.Permissives.AutoStart	Bool	☐	FALSE		
*DB_HMI".Status.Permissives.Valve[1]	Bool	☒	TRUE		
*DB_HMI".Status.Permissives.Valve[2]	Bool	☒	TRUE		
*DB_HMI".Status.Permissives.Valve[3]	Bool	☒	TRUE		
*DB_HMI".Status.Permissives.Pump[1]	Bool	☐	FALSE		
*DB_HMI".Status.Permissives.Pump[2]	Bool	☐	FALSE		
<Add new>					

TT-F10-A

Step 70 with all three tanks at 80 and all process commands off; no alarm/trip.

*FB_PlantController_DB".oHMStatus.Sequence.Active	Bool	☐	FALSE		
*FB_PlantController_DB".oHMStatus.Sequence.Complete	Bool	☒	TRUE		
*FB_PlantController_DB".oHMStatus.Sequence.Step	DEC+		70		
<Add new>					

TT-F10-B

Direct FB_PlantController_DB status: Sequence Step=70, Active=FALSE, Complete=TRUE.

Check Remove commands on operator Stop

Observed Stop returned the sequence to S0, disabled control and removed commands without a trip.

"DB_System" ScanCounter	DEC	2701725			01 PLC Health	
"DB_Simulation" Status.Tick100ms	Bool	☐ FALSE			01 PLC Health	
"DB_Config" Active.SimulationEnable	Bool	☒ TRUE			01 PLC Health	
"DB_HMI" Status.System.SimulationActive	Bool	☒ TRUE			01 PLC Health	
"DB_HMI" Status.ConfigValid	Bool	☒ TRUE			01 PLC Health	
"DB_HMI" Status.ConfigErrorCode	DEC	0			01 PLC Health	
"DB_HMI" Cmd.ModeRequest	DEC	2			02 Commands	0 Off, 1 Manual, 2 Auto
"DB_HMI" Cmd.Start	Bool	☐ FALSE			02 Commands	
"DB_HMI" Cmd.Stop	Bool	☐ FALSE			02 Commands	
"DB_HMI" Cmd.ResetFaults	Bool	☐ FALSE			02 Commands	
"DB_HMI" Cmd.ResetCycle	Bool	☐ FALSE			02 Commands	
"DB_HMI" Cmd.ResetSimulation	Bool	☐ FALSE	FALSE	☒ ⚠	02 Commands	
"DB_HMI" Status.System.Ready	Bool	☒ TRUE			03 System Status	
"DB_HMI" Status.System.ControlEnabled	Bool	☒ TRUE			03 System Status	
"DB_HMI" Status.System.ActiveMode	DEC	2			03 System Status	0 Off, 1 Manual, 2 Auto
"DB_HMI" Status.System.Offline	Bool	☐ FALSE			03 System Status	
"DB_HMI" Status.System.ManualActive	Bool	☐ FALSE			03 System Status	
"DB_HMI" Status.System.AutoActive	Bool	☒ TRUE			03 System Status	
"DB_HMI" Status.System.Faulted	Bool	☐ FALSE			03 System Status	
"DB_HMI" Status.Sequence.Step	DEC+-	20			04 Sequence	
"DB_HMI" Status.Sequence.Active	Bool	☐ FALSE			04 Sequence	
"DB_HMI" Status.Sequence.Complete	Bool	☐ FALSE			04 Sequence	
"DB_HMI" Status.Sequence.StepTimeout	Bool	☐ FALSE			04 Sequence	
"DB_HMI" Status.Sequence.TimeoutStep	DEC+-	0			04 Sequence	
"DB_HMI" Status.Sequence.ElapsedSec	Floating-point number	11.508			04 Sequence	
"DB_HMI" Status.Sequence.LastCycleTimeSec	Floating-point number	95.19912			04 Sequence	
"DB_HMI" Status.Sequence.CycleCounter	DEC	2			04 Sequence	
"DB_HMI" Status.Tank[1].LevelPct	Floating-point number	47.5			05 T101 Level	
"DB_HMI" Status.Tank[2].LevelPct	Floating-point number	0.0			05 T102 Level	
"DB_HMI" Status.Tank[3].LevelPct	Floating-point number	0.0			05 T103 Level	
"DB_IO" ProcessOut.ValveOpenCmd[1]	Bool	☒ TRUE			06 Process Commands	
"DB_IO" ProcessOut.ValveOpenCmd[2]	Bool	☐ FALSE			06 Process Commands	
"DB_IO" ProcessOut.ValveOpenCmd[3]	Bool	☐ FALSE			06 Process Commands	
"DB_IO" ProcessOut.PumpStartCmd[1]	Bool	☐ FALSE			06 Process Commands	
"DB_IO" ProcessOut.PumpStartCmd[2]	Bool	☐ FALSE			06 Process Commands	
"DB_IO" PhysicalOut.ValveOpenCmd[1]	Bool	☐ FALSE			07 Physical Output Gate	
"DB_IO" PhysicalOut.ValveOpenCmd[2]	Bool	☐ FALSE			07 Physical Output Gate	
"DB_IO" PhysicalOut.ValveOpenCmd[3]	Bool	☐ FALSE			07 Physical Output Gate	
"DB_IO" PhysicalOut.PumpStartCmd[1]	Bool	☐ FALSE			07 Physical Output Gate	
"DB_IO" PhysicalOut.PumpStartCmd[2]	Bool	☐ FALSE			07 Physical Output Gate	
"DB_Alarms" Alarm.ActiveWord	Hex	1680000_0000			08 Alarm Summary	
"DB_Alarms" Alarm.AnyAlarm	Bool	☐ FALSE			08 Alarm Summary	
"DB_Alarms" Alarm.AnyTrip	Bool	☐ FALSE			08 Alarm Summary	
"DB_Alarms" Alarm.FirstOutCode	DEC	0			08 Alarm Summary	
"PHL_DQ_XV101_OpenCmd"	%Q200.0 Bool	☐ FALSE				
"PHL_DQ_XV102_OpenCmd"	%Q200.1 Bool	☐ FALSE				
"PHL_DQ_XV103_OpenCmd"	%Q200.2 Bool	☐ FALSE				
"PHL_DQ_F102_StartCmd"	%Q200.3 Bool	☐ FALSE				
<Add new>						

TT-F11-A

Before STOP: Step 20, ControlEnabled TRUE, T101 about 47.5, XV101 command TRUE, no alarms/trips.

"DB_System" ScanCounter	DEC	2708255			01 PLC Health	
"DB_Simulation" Status.Tick100ms	Bool	☒ TRUE			01 PLC Health	
"DB_Config" Active.SimulationEnable	Bool	☒ TRUE			01 PLC Health	
"DB_HMI" Status.System.SimulationActive	Bool	☒ TRUE			01 PLC Health	
"DB_HMI" Status.ConfigValid	Bool	☒ TRUE			01 PLC Health	
"DB_HMI" Status.ConfigErrorCode	DEC	0			01 PLC Health	
"DB_HMI" Cmd.ModeRequest	DEC	2			02 Commands	0 Off, 1 Manual, 2 Auto
"DB_HMI" Cmd.Start	Bool	☐ FALSE			02 Commands	
"DB_HMI" Cmd.Stop	Bool	☐ FALSE			02 Commands	
"DB_HMI" Cmd.ResetFaults	Bool	☐ FALSE			02 Commands	
"DB_HMI" Cmd.ResetCycle	Bool	☐ FALSE			02 Commands	
"DB_HMI" Cmd.ResetSimulation	Bool	☐ FALSE	FALSE	☒ ⚠	02 Commands	
"DB_HMI" Status.System.Ready	Bool	☒ TRUE			03 System Status	
"DB_HMI" Status.System.ControlEnabled	Bool	☐ FALSE			03 System Status	
"DB_HMI" Status.System.ActiveMode	DEC	2			03 System Status	0 Off, 1 Manual, 2 Auto
"DB_HMI" Status.System.Offline	Bool	☐ FALSE			03 System Status	
"DB_HMI" Status.System.ManualActive	Bool	☐ FALSE			03 System Status	
"DB_HMI" Status.System.AutoActive	Bool	☒ TRUE			03 System Status	
"DB_HMI" Status.System.Faulted	Bool	☐ FALSE			03 System Status	
"DB_HMI" Status.Sequence.Step	DEC+-	0			04 Sequence	
"DB_HMI" Status.Sequence.Active	Bool	☐ FALSE			04 Sequence	
"DB_HMI" Status.Sequence.Complete	Bool	☐ FALSE			04 Sequence	
"DB_HMI" Status.Sequence.StepTimeout	Bool	☐ FALSE			04 Sequence	
"DB_HMI" Status.Sequence.TimeoutStep	DEC+-	0			04 Sequence	
"DB_HMI" Status.Sequence.ElapsedSec	Floating-point number	0.0			04 Sequence	
"DB_HMI" Status.Sequence.LastCycleTimeSec	Floating-point number	95.19912			04 Sequence	
"DB_HMI" Status.Sequence.CycleCounter	DEC	2			04 Sequence	
"DB_HMI" Status.Tank[1].LevelPct	Floating-point number	44.5			05 T101 Level	
"DB_HMI" Status.Tank[2].LevelPct	Floating-point number	0.0			05 T102 Level	
"DB_HMI" Status.Tank[3].LevelPct	Floating-point number	0.0			05 T103 Level	
"DB_IO" ProcessOut.ValveOpenCmd[1]	Bool	☐ FALSE			06 Process Commands	
"DB_IO" ProcessOut.ValveOpenCmd[2]	Bool	☐ FALSE			06 Process Commands	
"DB_IO" ProcessOut.ValveOpenCmd[3]	Bool	☐ FALSE			06 Process Commands	
"DB_IO" ProcessOut.PumpStartCmd[1]	Bool	☐ FALSE			06 Process Commands	
"DB_IO" ProcessOut.PumpStartCmd[2]	Bool	☐ FALSE			06 Process Commands	
"DB_IO" PhysicalOut.ValveOpenCmd[1]	Bool	☐ FALSE			07 Physical Output Gate	
"DB_IO" PhysicalOut.ValveOpenCmd[2]	Bool	☐ FALSE			07 Physical Output Gate	
"DB_IO" PhysicalOut.ValveOpenCmd[3]	Bool	☐ FALSE			07 Physical Output Gate	
"DB_IO" PhysicalOut.PumpStartCmd[1]	Bool	☐ FALSE			07 Physical Output Gate	
"DB_IO" PhysicalOut.PumpStartCmd[2]	Bool	☐ FALSE			07 Physical Output Gate	
"DB_Alarms" Alarm.ActiveWord	Hex	1680000_0000			08 Alarm Summary	
"DB_Alarms" Alarm.AnyAlarm	Bool	☐ FALSE			08 Alarm Summary	
"DB_Alarms" Alarm.AnyTrip	Bool	☐ FALSE			08 Alarm Summary	
"DB_Alarms" Alarm.FirstOutCode	DEC	0			08 Alarm Summary	
"PHL_DQ_XV101_OpenCmd"	%Q200.0 Bool	☐ FALSE				
"PHL_DQ_XV102_OpenCmd"	%Q200.1 Bool	☐ FALSE				
"PHL_DQ_XV103_OpenCmd"	%Q200.2 Bool	☐ FALSE				
"PHL_DQ_F102_StartCmd"	%Q200.3 Bool	☐ FALSE				
<Add new>						

TT-F11-B

After STOP: Step 0, ControlEnabled FALSE, all process commands off, no alarm/trip.

Check Stop during transfer

Observed Transfer commands were removed and the controller returned to S0 without a trip.

"DB_System" ScanCounter	DEC	2998143		01 PLC Health	
"DB_Simulation" Status.Tick100ms	Bool	☐ FALSE		01 PLC Health	
"DB_Config" Active.SimulationEnable	Bool	☒ TRUE		01 PLC Health	
"DB_HMI" Status.System.SimulationActive	Bool	☒ TRUE		01 PLC Health	
"DB_HMI" Status.ConfigValid	Bool	☒ TRUE		01 PLC Health	
"DB_HMI" Status.ConfigErrorCode	DEC	0		01 PLC Health	
"DB_HMI" Cmd.ModeRequest	DEC	2		02 Commands	0 Off, 1 Manual, 2 Auto
"DB_HMI" Cmd.Start	Bool	☐ FALSE		02 Commands	
"DB_HMI" Cmd.Stop	Bool	☐ FALSE		02 Commands	
"DB_HMI" Cmd.ResetFaults	Bool	☐ FALSE		02 Commands	
"DB_HMI" Cmd.ResetCycle	Bool	☐ FALSE		02 Commands	
"DB_HMI" Cmd.ResetSimulation	Bool	☐ FALSE		02 Commands	
"DB_HMI" Status.System.Ready	Bool	☒ TRUE		03 System Status	
"DB_HMI" Status.System.ControlEnabled	Bool	☒ TRUE		03 System Status	
"DB_HMI" Status.System.ActiveMode	DEC	2		03 System Status	0 Off, 1 Manual, 2 Auto
"DB_HMI" Status.System.Offline	Bool	☐ FALSE		03 System Status	
"DB_HMI" Status.System.ManualActive	Bool	☐ FALSE		03 System Status	
"DB_HMI" Status.System.AutoActive	Bool	☒ TRUE		03 System Status	
"DB_HMI" Status.System.Faulted	Bool	☐ FALSE		03 System Status	
"DB_HMI" Status.Sequence.Step	DEC+-	40		04 Sequence	
"DB_HMI" Status.Sequence.Active	Bool	☐ FALSE		04 Sequence	
"DB_HMI" Status.Sequence.Complete	Bool	☐ FALSE		04 Sequence	
"DB_HMI" Status.Sequence.StepTimeout	Bool	☐ FALSE		04 Sequence	
"DB_HMI" Status.Sequence.TimeoutStep	DEC+-	0		04 Sequence	
"DB_HMI" Status.Sequence.StepElapsedSec	Floating-point number	2.171		04 Sequence	
"DB_HMI" Status.Sequence.LastCycleTimeSec	Floating-point number	95.19912		04 Sequence	
"DB_HMI" Status.Sequence.CycleCounter	DEC	2		04 Sequence	
"DB_HMI" Status.Tank[1].LevelPct	Floating-point number	80.0		05 T101 Level	
"DB_HMI" Status.Tank[2].LevelPct	Floating-point number	0.0		05 T102 Level	
"DB_HMI" Status.Tank[3].LevelPct	Floating-point number	0.0		05 T103 Level	
"DB_IO" ProcessOut.ValveOpenCmd[1]	Bool	☐ FALSE		06 Process Commands	
"DB_IO" ProcessOut.ValveOpenCmd[2]	Bool	☒ TRUE		06 Process Commands	
"DB_IO" ProcessOut.ValveOpenCmd[3]	Bool	☐ FALSE		06 Process Commands	
"DB_IO" ProcessOut.PumpStartCmd[1]	Bool	☒ TRUE		06 Process Commands	
"DB_IO" ProcessOut.PumpStartCmd[2]	Bool	☐ FALSE		06 Process Commands	
"DB_IO" ProcessOut.PumpStartCmd[3]	Bool	☐ FALSE		06 Process Commands	
"DB_IO" PhysicalOut.ValveOpenCmd[1]	Bool	☐ FALSE		07 Physical Output Gate	
"DB_IO" PhysicalOut.ValveOpenCmd[2]	Bool	☐ FALSE		07 Physical Output Gate	
"DB_IO" PhysicalOut.ValveOpenCmd[3]	Bool	☐ FALSE		07 Physical Output Gate	
"DB_IO" PhysicalOut.PumpStartCmd[1]	Bool	☐ FALSE		07 Physical Output Gate	
"DB_IO" PhysicalOut.PumpStartCmd[2]	Bool	☐ FALSE		07 Physical Output Gate	
"DB_Alarms" Alarm.ActiveWord	Hex	1680000_0000		08 Alarm Summary	
"DB_Alarms" Alarm.AnyAlarm	Bool	☐ FALSE		08 Alarm Summary	
"DB_Alarms" Alarm.AnyTrip	Bool	☐ FALSE		08 Alarm Summary	
"DB_Alarms" Alarm.FirstOutCode	DEC	0		08 Alarm Summary	
"PHL_DQ_XV101_OpenCmd"	%Q200.0 Bool	☐ FALSE			
"PHL_DQ_XV102_OpenCmd"	%Q200.1 Bool	☐ FALSE			
"PHL_DQ_XV103_OpenCmd"	%Q200.2 Bool	☐ FALSE			
"PHL_DQ_P102_StartCmd"	%Q200.3 Bool	☐ FALSE			
<Add new>					

TT-F12-A

Before STOP: Step 40, XV102 and P101 process commands active, no trip.

"DB_System" ScanCounter	DEC	3015417		01 PLC Health	
"DB_Simulation" Status.Tick100ms	Bool	☐ FALSE		01 PLC Health	
"DB_Config" Active.SimulationEnable	Bool	☒ TRUE		01 PLC Health	
"DB_HMI" Status.System.SimulationActive	Bool	☒ TRUE		01 PLC Health	
"DB_HMI" Status.ConfigValid	Bool	☒ TRUE		01 PLC Health	
"DB_HMI" Status.ConfigErrorCode	DEC	0		01 PLC Health	
"DB_HMI" Cmd.ModeRequest	DEC	2		02 Commands	0 Off, 1 Manual, 2 Auto
"DB_HMI" Cmd.Start	Bool	☐ FALSE		02 Commands	
"DB_HMI" Cmd.Stop	Bool	☐ FALSE		02 Commands	
"DB_HMI" Cmd.ResetFaults	Bool	☐ FALSE		02 Commands	
"DB_HMI" Cmd.ResetCycle	Bool	☐ FALSE		02 Commands	
"DB_HMI" Cmd.ResetSimulation	Bool	☐ FALSE		02 Commands	
"DB_HMI" Status.System.Ready	Bool	☒ TRUE		03 System Status	
"DB_HMI" Status.System.ControlEnabled	Bool	☐ FALSE		03 System Status	
"DB_HMI" Status.System.ActiveMode	DEC	2		03 System Status	0 Off, 1 Manual, 2 Auto
"DB_HMI" Status.System.Offline	Bool	☐ FALSE		03 System Status	
"DB_HMI" Status.System.ManualActive	Bool	☐ FALSE		03 System Status	
"DB_HMI" Status.System.AutoActive	Bool	☒ TRUE		03 System Status	
"DB_HMI" Status.System.Faulted	Bool	☐ FALSE		03 System Status	
"DB_HMI" Status.Sequence.Step	DEC+-	0		04 Sequence	
"DB_HMI" Status.Sequence.Active	Bool	☐ FALSE		04 Sequence	
"DB_HMI" Status.Sequence.Complete	Bool	☐ FALSE		04 Sequence	
"DB_HMI" Status.Sequence.StepTimeout	Bool	☐ FALSE		04 Sequence	
"DB_HMI" Status.Sequence.TimeoutStep	DEC+-	0		04 Sequence	
"DB_HMI" Status.Sequence.StepElapsedSec	Floating-point number	0.0		04 Sequence	
"DB_HMI" Status.Sequence.LastCycleTimeSec	Floating-point number	95.19912		04 Sequence	
"DB_HMI" Status.Sequence.CycleCounter	DEC	2		04 Sequence	
"DB_HMI" Status.Tank[1].LevelPct	Floating-point number	69.0		05 T101 Level	
"DB_HMI" Status.Tank[2].LevelPct	Floating-point number	47.5		05 T102 Level	
"DB_HMI" Status.Tank[3].LevelPct	Floating-point number	0.0		05 T103 Level	
"DB_IO" ProcessOut.ValveOpenCmd[1]	Bool	☐ FALSE		06 Process Commands	
"DB_IO" ProcessOut.ValveOpenCmd[2]	Bool	☐ FALSE		06 Process Commands	
"DB_IO" ProcessOut.ValveOpenCmd[3]	Bool	☐ FALSE		06 Process Commands	
"DB_IO" ProcessOut.PumpStartCmd[1]	Bool	☐ FALSE		06 Process Commands	
"DB_IO" ProcessOut.PumpStartCmd[2]	Bool	☐ FALSE		06 Process Commands	
"DB_IO" PhysicalOut.ValveOpenCmd[1]	Bool	☐ FALSE		07 Physical Output Gate	
"DB_IO" PhysicalOut.ValveOpenCmd[2]	Bool	☐ FALSE		07 Physical Output Gate	
"DB_IO" PhysicalOut.ValveOpenCmd[3]	Bool	☐ FALSE		07 Physical Output Gate	
"DB_IO" PhysicalOut.PumpStartCmd[1]	Bool	☐ FALSE		07 Physical Output Gate	
"DB_IO" PhysicalOut.PumpStartCmd[2]	Bool	☐ FALSE		07 Physical Output Gate	
"DB_Alarms" Alarm.ActiveWord	Hex	1680000_0000		08 Alarm Summary	
"DB_Alarms" Alarm.AnyAlarm	Bool	☐ FALSE		08 Alarm Summary	
"DB_Alarms" Alarm.AnyTrip	Bool	☐ FALSE		08 Alarm Summary	
"DB_Alarms" Alarm.FirstOutCode	DEC	0		08 Alarm Summary	
"PHL_DQ_XV101_OpenCmd"	%Q200.0 Bool	☐ FALSE			
"PHL_DQ_XV102_OpenCmd"	%Q200.1 Bool	☐ FALSE			
"PHL_DQ_XV103_OpenCmd"	%Q200.2 Bool	☐ FALSE			
"PHL_DQ_P102_StartCmd"	%Q200.3 Bool	☐ FALSE			
<Add new>					

TT-F12-B

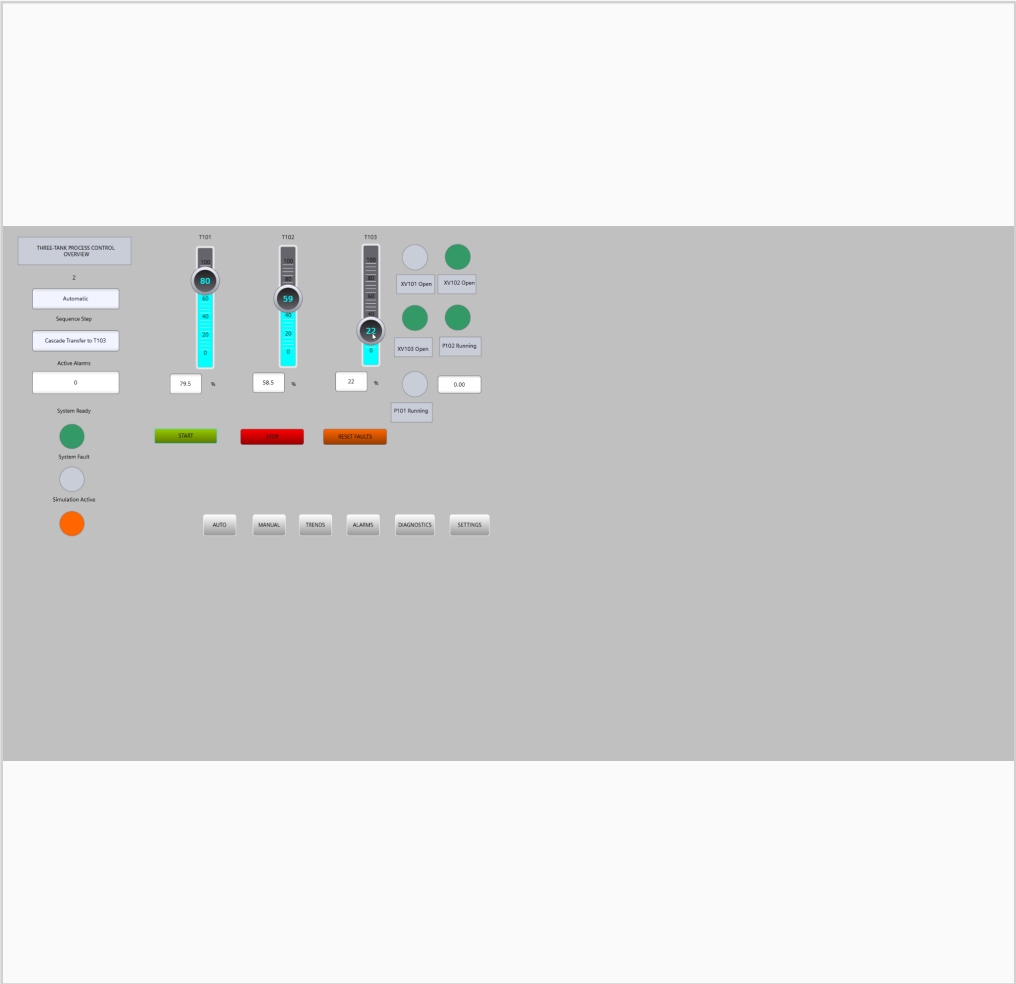
After STOP: Step 0, ControlEnabled FALSE, all process commands off, no alarm/trip.

Check Stop during cascade

Observed Cascade commands were removed and the controller returned to S0 without a trip.

"DB_System" ScanCounter	DEC	3145088		01 PLC Health	
"DB_Simulation" Status.Tick100ms	Bool	FALSE		01 PLC Health	
"DB_Config" Active.SimulationEnable	Bool	TRUE		01 PLC Health	
"DB_HMI" Status.System.SimulationActive	Bool	TRUE		01 PLC Health	
"DB_HMI" Status.ConfigValid	Bool	TRUE		01 PLC Health	
"DB_HMI" Status.ConfigErrorCode	DEC	0		01 PLC Health	
"DB_HMI" Cmd.ModeRequest	DEC	2		02 Commands	0 Off, 1 Manual, 2 Auto
"DB_HMI" Cmd.Start	Bool	FALSE		02 Commands	
"DB_HMI" Cmd.Stop	Bool	FALSE		02 Commands	
"DB_HMI" Cmd.ResetFaults	Bool	FALSE		02 Commands	
"DB_HMI" Cmd.ResetCycle	Bool	FALSE		02 Commands	
"DB_HMI" Cmd.ResetSimulation	Bool	FALSE		02 Commands	
"DB_HMI" Status.System.Ready	Bool	TRUE		03 System Status	
"DB_HMI" Status.System.ControlEnabled	Bool	TRUE		03 System Status	0 Off, 1 Manual, 2 Auto
"DB_HMI" Status.System.ActiveMode	DEC	2		03 System Status	
"DB_HMI" Status.System.Offline	Bool	FALSE		03 System Status	
"DB_HMI" Status.System.ManualActive	Bool	FALSE		03 System Status	
"DB_HMI" Status.System.AutoActive	Bool	TRUE		03 System Status	
"DB_HMI" Status.System.Faulted	Bool	FALSE		03 System Status	
"DB_HMI" Status.Sequence.Step	DEC	60		04 Sequence	
"DB_HMI" Status.Sequence.Active	Bool	FALSE		04 Sequence	
"DB_HMI" Status.Sequence.Complete	Bool	FALSE		04 Sequence	
"DB_HMI" Status.Sequence.StepTimeout	Bool	FALSE		04 Sequence	
"DB_HMI" Status.Sequence.TimeoutStep	DEC	0		04 Sequence	
"DB_HMI" Status.Sequence.StepElapsedSec	Floating-point number	5.731		04 Sequence	
"DB_HMI" Status.Sequence.LastCycleTimeSec	Floating-point number	95.19912		04 Sequence	
"DB_HMI" Status.Sequence.CycleCounter	DEC	2		04 Sequence	
"DB_HMI" Status.Tank[1].LevelPct	Floating-point number	80.0		05 T101 Level	
"DB_HMI" Status.Tank[2].LevelPct	Floating-point number	72.0		05 T102 Level	
"DB_HMI" Status.Tank[3].LevelPct	Floating-point number	8.0		05 T103 Level	
"DB_IO" ProcessOut.ValveOpenCmd[1]	Bool	FALSE		06 Process Commands	
"DB_IO" ProcessOut.ValveOpenCmd[2]	Bool	TRUE		06 Process Commands	
"DB_IO" ProcessOut.ValveOpenCmd[3]	Bool	TRUE		06 Process Commands	
"DB_IO" ProcessOut.PumpStartCmd[1]	Bool	FALSE		06 Process Commands	
"DB_IO" ProcessOut.PumpStartCmd[2]	Bool	TRUE		06 Process Commands	
"DB_IO" PhysicalOut.ValveOpenCmd[1]	Bool	FALSE		07 Physical Output Gate	
"DB_IO" PhysicalOut.ValveOpenCmd[2]	Bool	FALSE		07 Physical Output Gate	
"DB_IO" PhysicalOut.ValveOpenCmd[3]	Bool	FALSE		07 Physical Output Gate	
"DB_IO" PhysicalOut.PumpStartCmd[1]	Bool	FALSE		07 Physical Output Gate	
"DB_IO" PhysicalOut.PumpStartCmd[2]	Bool	FALSE		07 Physical Output Gate	
"DB_Alarms" Alarm.ActiveWord	Hex	1680000_0000		08 Alarm Summary	
"DB_Alarms" Alarm.AnyAlarm	Bool	FALSE		08 Alarm Summary	
"DB_Alarms" Alarm.AnyTrip	Bool	FALSE		08 Alarm Summary	
"DB_Alarms" Alarm.FirstOutCode	DEC	0		08 Alarm Summary	
"PH_DQ_XV101_OpenCmd"	%Q200.0	Bool	FALSE		
"PH_DQ_XV102_OpenCmd"	%Q200.1	Bool	FALSE		
"PH_DQ_XV103_OpenCmd"	%Q200.2	Bool	FALSE		
"PH_DQ_P102_StartCmd"	%Q200.3	Bool	FALSE		
<Add new>					

TT-F13-A
Before STOP: Step 60, T101=80, T102=72, T103=8; XV102/XV103 and P102 process commands active; no trip.



TT-F13-B
WinCC Overview showing Cascade Transfer to T103 with T103 rising.

Check Trip on missing XV101 proof

Observed Valve 1 fault code 1, S90 and first-out 1401 were recorded.

"DB_System" ScanCounter	DEC	4048686			01 PLC Health	
"DB_Simulation" Status.Tick100ms	Bool	☐ FALSE			01 PLC Health	
"DB_Config" Active.SimulationEnable	Bool	☒ TRUE			01 PLC Health	
"DB_HMA" Status.System.SimulationActive	Bool	☒ TRUE			01 PLC Health	
"DB_HMA" Status.ConfigValid	Bool	☒ TRUE			01 PLC Health	
"DB_HMA" Status.ConfigErrorCode	DEC	0			01 PLC Health	
"DB_HMA" Cmd.ModeRequest	DEC	2			02 Commands	0 Off, 1 Manual, 2 Auto
"DB_HMA" Cmd.Start	Bool	☐ FALSE	FALSE	☒ TRUE	02 Commands	
"DB_HMA" Cmd.Stop	Bool	☐ FALSE			02 Commands	
"DB_HMA" Cmd.ResetFaults	Bool	☐ FALSE			02 Commands	
"DB_HMA" Cmd.ResetCycle	Bool	☐ FALSE			02 Commands	
"DB_HMA" Cmd.ResetSimulation	Bool	☐ FALSE			02 Commands	
"DB_HMA" Status.System.Ready	Bool	☒ TRUE			03 System Status	
"DB_HMA" Status.System.ControlEnabled	Bool	☒ TRUE			03 System Status	
"DB_HMA" Status.System.ActiveMode	DEC	2			03 System Status	0 Off, 1 Manual, 2 Auto
"DB_HMA" Status.System.Offline	Bool	☐ FALSE			03 System Status	
"DB_HMA" Status.System.ManualActive	Bool	☐ FALSE			03 System Status	
"DB_HMA" Status.System.AutoActive	Bool	☒ TRUE			03 System Status	
"DB_HMA" Status.System.Faulted	Bool	☐ FALSE			03 System Status	
"DB_HMA" Status.Sequence.Step	DEC+-	10			04 Sequence	
"DB_HMA" Status.Sequence.Active	Bool	☐ FALSE			04 Sequence	
"DB_HMA" Status.Sequence.Complete	Bool	☐ FALSE			04 Sequence	
"DB_HMA" Status.Sequence.StepTimeout	Bool	☐ FALSE			04 Sequence	
"DB_HMA" Status.Sequence.TimeoutStep	DEC+-	0			04 Sequence	
"DB_HMA" Status.Sequence.StepElapsedSec	Floating-point number	3.465			04 Sequence	
"DB_HMA" Status.Sequence.LastCycleTimeSec	Floating-point number	95.19912			04 Sequence	
"DB_HMA" Status.Sequence.CycleCounter	DEC	2			04 Sequence	
"DB_HMA" Status.Tank[1].LevelPct	Floating-point number	0.0			05 TI01 Level	
"DB_HMA" Status.Tank[2].LevelPct	Floating-point number	0.0			05 TI02 Level	
"DB_HMA" Status.Tank[3].LevelPct	Floating-point number	0.0			05 TI03 Level	
"DB_IO" ProcessOut.ValveOpenCmd[1]	Bool	☐ FALSE			06 Process Commands	
"DB_IO" ProcessOut.ValveOpenCmd[2]	Bool	☐ FALSE			06 Process Commands	
"DB_IO" ProcessOut.ValveOpenCmd[3]	Bool	☐ FALSE			06 Process Commands	
"DB_IO" ProcessOut.PumpStartCmd[1]	Bool	☐ FALSE			06 Process Commands	
"DB_IO" ProcessOut.PumpStartCmd[2]	Bool	☐ FALSE			06 Process Commands	
"DB_IO" PhysicalOut.ValveOpenCmd[1]	Bool	☐ FALSE			07 Physical Output Gate	
"DB_IO" PhysicalOut.ValveOpenCmd[2]	Bool	☐ FALSE			07 Physical Output Gate	
"DB_IO" PhysicalOut.ValveOpenCmd[3]	Bool	☐ FALSE			07 Physical Output Gate	
"DB_IO" PhysicalOut.PumpStartCmd[1]	Bool	☐ FALSE			07 Physical Output Gate	
"DB_IO" PhysicalOut.PumpStartCmd[2]	Bool	☐ FALSE			07 Physical Output Gate	
"DB_Alarms" Alarm.ActiveWord	Hex	1680000_0000			08 Alarm Summary	
"DB_Alarms" Alarm.AnyAlarm	Bool	☐ FALSE			08 Alarm Summary	
"DB_Alarms" Alarm.AnyTrip	Bool	☐ FALSE			08 Alarm Summary	
"DB_Alarms" Alarm.FirstOutCode	DEC	0			08 Alarm Summary	
"PHL_DQ_XV101_OpenCmd"	%Q200.0	☐ FALSE				
"PHL_DQ_XV102_OpenCmd"	%Q200.1	☐ FALSE				
"PHL_DQ_XV103_OpenCmd"	%Q200.2	☐ FALSE				
"PHL_DQ_F102_StartCmd"	%Q200.3	☐ FALSE				
"DB_HMA" Status.Valve[1].Fault	Bool	☐ FALSE				
"DB_HMA" Status.Valve[1].FaultCode	DEC	0				
"DB_Simulation" Inject.FaultValveOpenPh[1]	Bool	☒ TRUE	TRUE	☒ TRUE		

TT-F14-A

XV101 feedback-failure injection active while sequence is entering normal operation; no fault yet.

"DB_System" ScanCounter	DEC	4060673			01 PLC Health	
"DB_Simulation" Status.Tick100ms	Bool	☒ TRUE			01 PLC Health	
"DB_Config" Active.SimulationEnable	Bool	☒ TRUE			01 PLC Health	
"DB_HMA" Status.System.SimulationActive	Bool	☒ TRUE			01 PLC Health	
"DB_HMA" Status.ConfigValid	Bool	☒ TRUE			01 PLC Health	
"DB_HMA" Status.ConfigErrorCode	DEC	0			01 PLC Health	
"DB_HMA" Cmd.ModeRequest	DEC	2			02 Commands	0 Off, 1 Manual, 2 Auto
"DB_HMA" Cmd.Start	Bool	☐ FALSE	FALSE	☒ TRUE	02 Commands	
"DB_HMA" Cmd.Stop	Bool	☐ FALSE			02 Commands	
"DB_HMA" Cmd.ResetFaults	Bool	☐ FALSE			02 Commands	
"DB_HMA" Cmd.ResetCycle	Bool	☐ FALSE			02 Commands	
"DB_HMA" Cmd.ResetSimulation	Bool	☐ FALSE			02 Commands	
"DB_HMA" Status.System.Ready	Bool	☐ FALSE			03 System Status	
"DB_HMA" Status.System.ControlEnabled	Bool	☐ FALSE			03 System Status	
"DB_HMA" Status.System.ActiveMode	DEC	2			03 System Status	0 Off, 1 Manual, 2 Auto
"DB_HMA" Status.System.Offline	Bool	☐ FALSE			03 System Status	
"DB_HMA" Status.System.ManualActive	Bool	☐ FALSE			03 System Status	
"DB_HMA" Status.System.AutoActive	Bool	☒ TRUE			03 System Status	
"DB_HMA" Status.System.Faulted	Bool	☒ TRUE			03 System Status	
"DB_HMA" Status.Sequence.Step	DEC+-	90			04 Sequence	
"DB_HMA" Status.Sequence.Active	Bool	☐ FALSE			04 Sequence	
"DB_HMA" Status.Sequence.Complete	Bool	☐ FALSE			04 Sequence	
"DB_HMA" Status.Sequence.StepTimeout	Bool	☐ FALSE			04 Sequence	
"DB_HMA" Status.Sequence.TimeoutStep	DEC+-	0			04 Sequence	
"DB_HMA" Status.Sequence.StepElapsedSec	Floating-point number	0.0			04 Sequence	
"DB_HMA" Status.Sequence.LastCycleTimeSec	Floating-point number	95.19912			04 Sequence	
"DB_HMA" Status.Sequence.CycleCounter	DEC	2			04 Sequence	
"DB_HMA" Status.Tank[1].LevelPct	Floating-point number	0.0			05 TI01 Level	
"DB_HMA" Status.Tank[2].LevelPct	Floating-point number	0.0			05 TI02 Level	
"DB_HMA" Status.Tank[3].LevelPct	Floating-point number	0.0			05 TI03 Level	
"DB_IO" ProcessOut.ValveOpenCmd[1]	Bool	☐ FALSE			06 Process Commands	
"DB_IO" ProcessOut.ValveOpenCmd[2]	Bool	☐ FALSE			06 Process Commands	
"DB_IO" ProcessOut.ValveOpenCmd[3]	Bool	☐ FALSE			06 Process Commands	
"DB_IO" ProcessOut.PumpStartCmd[1]	Bool	☐ FALSE			06 Process Commands	
"DB_IO" ProcessOut.PumpStartCmd[2]	Bool	☐ FALSE			06 Process Commands	
"DB_IO" PhysicalOut.ValveOpenCmd[1]	Bool	☐ FALSE			07 Physical Output Gate	
"DB_IO" PhysicalOut.ValveOpenCmd[2]	Bool	☐ FALSE			07 Physical Output Gate	
"DB_IO" PhysicalOut.ValveOpenCmd[3]	Bool	☐ FALSE			07 Physical Output Gate	
"DB_IO" PhysicalOut.PumpStartCmd[1]	Bool	☐ FALSE			07 Physical Output Gate	
"DB_IO" PhysicalOut.PumpStartCmd[2]	Bool	☐ FALSE			07 Physical Output Gate	
"DB_Alarms" Alarm.ActiveWord	Hex	1680000_1000			08 Alarm Summary	
"DB_Alarms" Alarm.AnyAlarm	Bool	☒ TRUE			08 Alarm Summary	
"DB_Alarms" Alarm.AnyTrip	Bool	☒ TRUE			08 Alarm Summary	
"DB_Alarms" Alarm.FirstOutCode	DEC	1401			08 Alarm Summary	
"PHL_DQ_XV101_OpenCmd"	%Q200.0	☐ FALSE				
"PHL_DQ_XV102_OpenCmd"	%Q200.1	☐ FALSE				
"PHL_DQ_XV103_OpenCmd"	%Q200.2	☐ FALSE				
"PHL_DQ_F102_StartCmd"	%Q200.3	☐ FALSE				
"DB_HMA" Status.Valve[1].Fault	Bool	☒ TRUE				
"DB_HMA" Status.Valve[1].FaultCode	DEC	1				
"DB_Simulation" Inject.FaultValveOpenPh[1]	Bool	☒ TRUE	TRUE	☒ TRUE		

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Step 90; System.Faulted TRUE; Valve[1].Fault TRUE / FaultCode 1; AnyAlarm/AnyTrip TRUE; FirstOutCode 1401; outputs removed.

Check Block P101 and trip on missing XV102 proof

Observed P101 remained off before proof; timeout produced valve 2 fault code 1, S90 and first-out 1402.

"DB_System" ScanCounter	DEC	5798066			01 PLC Health		
"DB_Simulation" Status.Tick100ms	Bool	TRUE			01 PLC Health		
"DB_Config" Active.SimulationEnable	Bool	TRUE			01 PLC Health		
"DB_HMA" Status.System.SimulationActive	Bool	TRUE			01 PLC Health		
"DB_HMA" Status.ConfigValid	Bool	TRUE			01 PLC Health		
"DB_HMA" Status.ConfigErrorCode	DEC	0			01 PLC Health		
"DB_HMA" Cmd.ModeRequest	DEC	2			02 Commands	0 Off, 1 Manual, 2 Auto	
"DB_HMA" Cmd.Start	Bool	FALSE			02 Commands		
"DB_HMA" Cmd.Stop	Bool	FALSE			02 Commands		
"DB_HMA" Cmd.ResetFaults	Bool	FALSE			02 Commands		
"DB_HMA" Cmd.ResetCycle	Bool	FALSE			02 Commands		
"DB_HMA" Cmd.ResetSimulation	Bool	FALSE			02 Commands		
"DB_HMA" Status.System.Ready	Bool	TRUE			03 System Status		
"DB_HMA" Status.System.ControlEnabled	Bool	TRUE			03 System Status		
"DB_HMA" Status.System.ActiveMode	DEC	2			03 System Status	0 Off, 1 Manual, 2 Auto	
"DB_HMA" Status.System.Offline	Bool	FALSE			03 System Status		
"DB_HMA" Status.System.ManualActive	Bool	FALSE			03 System Status		
"DB_HMA" Status.System.AutoActive	Bool	TRUE			03 System Status		
"DB_HMA" Status.System.Faulted	Bool	FALSE			03 System Status		
"DB_HMA" Status.Sequence.Step	DEC	40			04 Sequence		
"DB_HMA" Status.Sequence.Active	Bool	FALSE			04 Sequence		
"DB_HMA" Status.Sequence.Complete	Bool	FALSE			04 Sequence		
"DB_HMA" Status.Sequence.StepTimeout	Bool	FALSE			04 Sequence		
"DB_HMA" Status.Sequence.TimeoutStep	DEC	0			04 Sequence		
"DB_HMA" Status.Sequence.StepElapsedSec	Floating-point number	0.616			04 Sequence		
"DB_HMA" Status.Sequence.LastCycleTimeSec	Floating-point number	95.19912			04 Sequence		
"DB_HMA" Status.Sequence.CycleCounter	DEC	2			04 Sequence		
"DB_HMA" Status.Tank[1].LevelPct	Floating-point number	80.0			05 T101 Level		
"DB_HMA" Status.Tank[2].LevelPct	Floating-point number	0.0			05 T102 Level		
"DB_HMA" Status.Tank[3].LevelPct	Floating-point number	0.0			05 T103 Level		
"DB_IO" ProcessOut.ValveOpenCmd[1]	Bool	FALSE			06 Process Commands		
"DB_IO" ProcessOut.ValveOpenCmd[2]	Bool	TRUE			06 Process Commands		
"DB_IO" ProcessOut.ValveOpenCmd[3]	Bool	FALSE			06 Process Commands		
"DB_IO" ProcessOut.PumpStartCmd[1]	Bool	FALSE			06 Process Commands		
"DB_IO" ProcessOut.PumpStartCmd[2]	Bool	FALSE			06 Process Commands		
"DB_IO" PhysicalOut.ValveOpenCmd[1]	Bool	FALSE			07 Physical Output Gate		
"DB_IO" PhysicalOut.ValveOpenCmd[2]	Bool	FALSE			07 Physical Output Gate		
"DB_IO" PhysicalOut.ValveOpenCmd[3]	Bool	FALSE			07 Physical Output Gate		
"DB_IO" PhysicalOut.PumpStartCmd[1]	Bool	FALSE			07 Physical Output Gate		
"DB_IO" PhysicalOut.PumpStartCmd[2]	Bool	FALSE			07 Physical Output Gate		
"DB_Alarms" Alarm.ActiveWord	Hex	1680000_0000			08 Alarm Summary		
"DB_Alarms" Alarm.AnyAlarm	Bool	FALSE			08 Alarm Summary		
"DB_Alarms" Alarm.AnyTrip	Bool	FALSE			08 Alarm Summary		
"DB_Alarms" Alarm.FirstOutCode	DEC	0			08 Alarm Summary		
"PHL_DQ_XV101_OpenCmd"	%Q200.0	Bool	FALSE				
"PHL_DQ_XV102_OpenCmd"	%Q200.1	Bool	FALSE				
"PHL_DQ_XV103_OpenCmd"	%Q200.2	Bool	FALSE				
"PHL_DQ_P102_StartCmd"	%Q200.3	Bool	FALSE				
"DB_HMA" Status.Valve[1].Fault	DEC	0					
"DB_Simulation" Inject.FailValveOpenFb[1]	Bool	FALSE					
"DB_Simulation" Inject.FailValveOpenFb[2]	Bool	TRUE	TRUE				
"DB_HMA" Status.Valve[2].Fault	Bool	FALSE					
"DB_HMA" Status.Valve[2].FaultCode	DEC	0					
"DB_IO" SelectedIn.Valve[2].OpenFb	Bool	FALSE					
"DB_IO" SelectedIn.Pump[1].RunFb	Bool	FALSE					

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Step 40 with FailValveOpenFb[2]=TRUE; XV102 command TRUE, OpenFb FALSE, P101 command/run feedback FALSE.

"DB_System" ScanCounter	DEC	5803607			01 PLC Health		
"DB_Simulation" Status.Tick100ms	Bool	FALSE			01 PLC Health		
"DB_Config" Active.SimulationEnable	Bool	TRUE			01 PLC Health		
"DB_HMA" Status.System.SimulationActive	Bool	TRUE			01 PLC Health		
"DB_HMA" Status.ConfigValid	Bool	TRUE			01 PLC Health		
"DB_HMA" Status.ConfigErrorCode	DEC	0			01 PLC Health		
"DB_HMA" Cmd.ModeRequest	DEC	2			02 Commands	0 Off, 1 Manual, 2 Auto	
"DB_HMA" Cmd.Start	Bool	FALSE			02 Commands		
"DB_HMA" Cmd.Stop	Bool	FALSE			02 Commands		
"DB_HMA" Cmd.ResetFaults	Bool	FALSE			02 Commands		
"DB_HMA" Cmd.ResetCycle	Bool	FALSE			02 Commands		
"DB_HMA" Cmd.ResetSimulation	Bool	FALSE			02 Commands		
"DB_HMA" Status.System.Ready	Bool	FALSE			03 System Status		
"DB_HMA" Status.System.ControlEnabled	Bool	FALSE			03 System Status		
"DB_HMA" Status.System.ActiveMode	DEC	2			03 System Status	0 Off, 1 Manual, 2 Auto	
"DB_HMA" Status.System.Offline	Bool	FALSE			03 System Status		
"DB_HMA" Status.System.ManualActive	Bool	FALSE			03 System Status		
"DB_HMA" Status.System.AutoActive	Bool	TRUE			03 System Status		
"DB_HMA" Status.System.Faulted	Bool	TRUE			03 System Status		
"DB_HMA" Status.Sequence.Step	DEC	90			04 Sequence		
"DB_HMA" Status.Sequence.Active	Bool	FALSE			04 Sequence		
"DB_HMA" Status.Sequence.Complete	Bool	FALSE			04 Sequence		
"DB_HMA" Status.Sequence.StepTimeout	Bool	FALSE			04 Sequence		
"DB_HMA" Status.Sequence.TimeoutStep	DEC	0			04 Sequence		
"DB_HMA" Status.Sequence.StepElapsedSec	Floating-point number	0.0			04 Sequence		
"DB_HMA" Status.Sequence.LastCycleTimeSec	Floating-point number	95.19912			04 Sequence		
"DB_HMA" Status.Sequence.CycleCounter	DEC	2			04 Sequence		
"DB_HMA" Status.Tank[1].LevelPct	Floating-point number	80.0			05 T101 Level		
"DB_HMA" Status.Tank[2].LevelPct	Floating-point number	0.0			05 T102 Level		
"DB_HMA" Status.Tank[3].LevelPct	Floating-point number	0.0			05 T103 Level		
"DB_IO" ProcessOut.ValveOpenCmd[1]	Bool	FALSE			06 Process Commands		
"DB_IO" ProcessOut.ValveOpenCmd[2]	Bool	FALSE			06 Process Commands		
"DB_IO" ProcessOut.ValveOpenCmd[3]	Bool	FALSE			06 Process Commands		
"DB_IO" ProcessOut.PumpStartCmd[1]	Bool	FALSE			06 Process Commands		
"DB_IO" ProcessOut.PumpStartCmd[2]	Bool	FALSE			06 Process Commands		
"DB_IO" PhysicalOut.ValveOpenCmd[1]	Bool	FALSE			07 Physical Output Gate		
"DB_IO" PhysicalOut.ValveOpenCmd[2]	Bool	FALSE			07 Physical Output Gate		
"DB_IO" PhysicalOut.ValveOpenCmd[3]	Bool	FALSE			07 Physical Output Gate		
"DB_IO" PhysicalOut.PumpStartCmd[1]	Bool	FALSE			07 Physical Output Gate		
"DB_IO" PhysicalOut.PumpStartCmd[2]	Bool	FALSE			07 Physical Output Gate		
"DB_Alarms" Alarm.ActiveWord	Hex	1680000_2000			08 Alarm Summary		
"DB_Alarms" Alarm.AnyAlarm	Bool	TRUE			08 Alarm Summary		
"DB_Alarms" Alarm.AnyTrip	Bool	TRUE			08 Alarm Summary		
"DB_Alarms" Alarm.FirstOutCode	DEC	1402			08 Alarm Summary		
"PHL_DQ_XV101_OpenCmd"	%Q200.0	Bool	FALSE				
"PHL_DQ_XV102_OpenCmd"	%Q200.1	Bool	FALSE				
"PHL_DQ_XV103_OpenCmd"	%Q200.2	Bool	FALSE				
"PHL_DQ_P102_StartCmd"	%Q200.3	Bool	FALSE				
"DB_HMA" Status.Valve[1].Fault	Bool	FALSE					
"DB_HMA" Status.Valve[1].FaultCode	DEC	0					
"DB_Simulation" Inject.FailValveOpenFb[1]	Bool	FALSE					
"DB_Simulation" Inject.FailValveOpenFb[2]	Bool	TRUE	TRUE				
"DB_HMA" Status.Valve[2].Fault	Bool	TRUE					
"DB_HMA" Status.Valve[2].FaultCode	DEC	1					
"DB_IO" SelectedIn.Valve[2].OpenFb	Bool	FALSE					
"DB_IO" SelectedIn.Pump[1].RunFb	Bool	FALSE					

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Step 90; Valve[2].Fault TRUE / FaultCode 1; AnyAlarm/AnyTrip TRUE; FirstOut 1402; commands removed.

Check Block P102 and trip on missing XV103 proof

Observed P102 remained off before proof; timeout produced valve 3 fault code 1, S90 and first-out 1403.

"DB_System" ScanCounter	DEC	6044913			01 PLC Health	
"DB_Simulation" Status.Tick100ms	Bool	TRUE			01 PLC Health	
"DB_Config" Active.SimulationEnable	Bool	TRUE			01 PLC Health	
"DB_HMA" Status.System.SimulationActive	Bool	TRUE			01 PLC Health	
"DB_HMA" Status.ConfigValid	Bool	TRUE			01 PLC Health	
"DB_HMA" Status.ConfigErrorCode	DEC	0			01 PLC Health	
"DB_HMA" Cmd.ModeRequest	DEC	2			02 Commands	0 Off, 1 Manual, 2 Auto
"DB_HMA" Cmd.Start	Bool	FALSE			02 Commands	
"DB_HMA" Cmd.Stop	Bool	FALSE			02 Commands	
"DB_HMA" Cmd.ResetFaults	Bool	FALSE			02 Commands	
"DB_HMA" Cmd.ResetCycle	Bool	FALSE	FALSE		02 Commands	
"DB_HMA" Cmd.ResetSimulation	Bool	FALSE	FALSE		02 Commands	
"DB_HMA" Status.System.Ready	Bool	TRUE			03 System Status	
"DB_HMA" Status.System.ControlEnabled	Bool	TRUE			03 System Status	
"DB_HMA" Status.System.ActiveMode	DEC	2			03 System Status	0 Off, 1 Manual, 2 Auto
"DB_HMA" Status.System.Offline	Bool	FALSE			03 System Status	
"DB_HMA" Status.System.ManualActive	Bool	FALSE			03 System Status	
"DB_HMA" Status.System.AutoActive	Bool	TRUE			03 System Status	
"DB_HMA" Status.System.Faulted	Bool	FALSE			03 System Status	
"DB_HMA" Status.Sequence.Step	DEC+-	60			04 Sequence	
"DB_HMA" Status.Sequence.Active	Bool	FALSE			04 Sequence	
"DB_HMA" Status.Sequence.Complete	Bool	FALSE			04 Sequence	
"DB_HMA" Status.Sequence.StepTimeout	Bool	FALSE			04 Sequence	
"DB_HMA" Status.Sequence.TimeoutStep	DEC+-	0			04 Sequence	
"DB_HMA" Status.Sequence.StepElapsedSec	Floating-point number	0.536			04 Sequence	
"DB_HMA" Status.Sequence.LastCycleTimeSec	Floating-point number	95.19912			04 Sequence	
"DB_HMA" Status.Sequence.CycleCounter	DEC	2			04 Sequence	
"DB_HMA" Status.Tank[1].LevelPct	Floating-point number	80.0			05 T101 Level	
"DB_HMA" Status.Tank[2].LevelPct	Floating-point number	80.0			05 T102 Level	
"DB_HMA" Status.Tank[3].LevelPct	Floating-point number	0.0			05 T103 Level	
"DB_IO" ProcessOut.ValveOpenCmd[1]	Bool	FALSE			06 Process Commands	
"DB_IO" ProcessOut.ValveOpenCmd[2]	Bool	FALSE			06 Process Commands	
"DB_IO" ProcessOut.ValveOpenCmd[3]	Bool	TRUE			06 Process Commands	
"DB_IO" ProcessOut.PumpStartCmd[1]	Bool	FALSE			06 Process Commands	
"DB_IO" ProcessOut.PumpStartCmd[2]	Bool	FALSE			06 Process Commands	
"DB_IO" PhysicalOut.ValveOpenCmd[1]	Bool	FALSE			07 Physical Output Gate	
"DB_IO" PhysicalOut.ValveOpenCmd[2]	Bool	FALSE			07 Physical Output Gate	
"DB_IO" PhysicalOut.ValveOpenCmd[3]	Bool	FALSE			07 Physical Output Gate	
"DB_IO" PhysicalOut.PumpStartCmd[1]	Bool	FALSE			07 Physical Output Gate	
"DB_IO" PhysicalOut.PumpStartCmd[2]	Bool	FALSE			07 Physical Output Gate	
"DB_Alarms" Alarm.ActiveWord	Hex	1680000_0000			08 Alarm Summary	
"DB_Alarms" Alarm.AnyAlarm	Bool	FALSE			08 Alarm Summary	
"DB_Alarms" Alarm.AnyTrip	Bool	FALSE			08 Alarm Summary	
"DB_Alarms" Alarm.FirstOutCode	DEC	0			08 Alarm Summary	
"PHL_DQ_XV101_OpenCmd"	%Q200.0	Bool	FALSE			
"PHL_DQ_XV102_OpenCmd"	%Q200.1	Bool	FALSE			
"PHL_DQ_XV103_OpenCmd"	%Q200.2	Bool	FALSE			
"PHL_DQ_P102_StartCmd"	%Q200.3	Bool	FALSE			
"DB_HMA" Status.Valve[1].Fault	Bool	FALSE				
"DB_Simulation" Inject.FailValveOpenFb[1]	Bool	FALSE				
"DB_Simulation" Inject.FailValveOpenFb[2]	Bool	FALSE				
"DB_Simulation" Inject.FailValveOpenFb[3]	Bool	TRUE	TRUE			
"DB_HMA" Status.Valve[3].Fault	Bool	FALSE				
"DB_HMA" Status.Valve[3].FaultCode	DEC	0				
"DB_IO" SelectedIn.Valve[3].OpenFb	Bool	FALSE				
"DB_IO" SelectedIn.Pump[2].RunFb	Bool	FALSE				

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Step 60 with FailValveOpenFb[3]=TRUE; XV103 command active, OpenFb FALSE, P102 command/run feedback FALSE.

"DB_System" ScanCounter	DEC	6050257			01 PLC Health	
"DB_Simulation" Status.Tick100ms	Bool	TRUE			01 PLC Health	
"DB_Config" Active.SimulationEnable	Bool	TRUE			01 PLC Health	
"DB_HMA" Status.System.SimulationActive	Bool	TRUE			01 PLC Health	
"DB_HMA" Status.ConfigValid	Bool	TRUE			01 PLC Health	
"DB_HMA" Status.ConfigErrorCode	DEC	0			01 PLC Health	
"DB_HMA" Cmd.ModeRequest	DEC	2			02 Commands	0 Off, 1 Manual, 2 Auto
"DB_HMA" Cmd.Start	Bool	FALSE			02 Commands	
"DB_HMA" Cmd.Stop	Bool	FALSE			02 Commands	
"DB_HMA" Cmd.ResetFaults	Bool	FALSE			02 Commands	
"DB_HMA" Cmd.ResetCycle	Bool	FALSE	FALSE		02 Commands	
"DB_HMA" Cmd.ResetSimulation	Bool	FALSE	FALSE		02 Commands	
"DB_HMA" Status.System.Ready	Bool	FALSE			03 System Status	
"DB_HMA" Status.System.ControlEnabled	Bool	FALSE			03 System Status	
"DB_HMA" Status.System.ActiveMode	DEC	2			03 System Status	0 Off, 1 Manual, 2 Auto
"DB_HMA" Status.System.Offline	Bool	FALSE			03 System Status	
"DB_HMA" Status.System.ManualActive	Bool	FALSE			03 System Status	
"DB_HMA" Status.System.AutoActive	Bool	TRUE			03 System Status	
"DB_HMA" Status.System.Faulted	Bool	TRUE			03 System Status	
"DB_HMA" Status.Sequence.Step	DEC+-	90			04 Sequence	
"DB_HMA" Status.Sequence.Active	Bool	FALSE			04 Sequence	
"DB_HMA" Status.Sequence.Complete	Bool	FALSE			04 Sequence	
"DB_HMA" Status.Sequence.StepTimeout	Bool	FALSE			04 Sequence	
"DB_HMA" Status.Sequence.TimeoutStep	DEC+-	0			04 Sequence	
"DB_HMA" Status.Sequence.StepElapsedSec	Floating-point number	0.0			04 Sequence	
"DB_HMA" Status.Sequence.LastCycleTimeSec	Floating-point number	95.19912			04 Sequence	
"DB_HMA" Status.Sequence.CycleCounter	DEC	2			04 Sequence	
"DB_HMA" Status.Tank[1].LevelPct	Floating-point number	80.0			05 T101 Level	
"DB_HMA" Status.Tank[2].LevelPct	Floating-point number	80.0			05 T102 Level	
"DB_HMA" Status.Tank[3].LevelPct	Floating-point number	0.0			05 T103 Level	
"DB_IO" ProcessOut.ValveOpenCmd[1]	Bool	FALSE			06 Process Commands	
"DB_IO" ProcessOut.ValveOpenCmd[2]	Bool	FALSE			06 Process Commands	
"DB_IO" ProcessOut.ValveOpenCmd[3]	Bool	FALSE			06 Process Commands	
"DB_IO" ProcessOut.PumpStartCmd[1]	Bool	FALSE			06 Process Commands	
"DB_IO" ProcessOut.PumpStartCmd[2]	Bool	FALSE			06 Process Commands	
"DB_IO" PhysicalOut.ValveOpenCmd[1]	Bool	FALSE			07 Physical Output Gate	
"DB_IO" PhysicalOut.ValveOpenCmd[2]	Bool	FALSE			07 Physical Output Gate	
"DB_IO" PhysicalOut.ValveOpenCmd[3]	Bool	FALSE			07 Physical Output Gate	
"DB_IO" PhysicalOut.PumpStartCmd[1]	Bool	FALSE			07 Physical Output Gate	
"DB_IO" PhysicalOut.PumpStartCmd[2]	Bool	FALSE			07 Physical Output Gate	
"DB_Alarms" Alarm.ActiveWord	Hex	1680000_4000			08 Alarm Summary	
"DB_Alarms" Alarm.AnyAlarm	Bool	TRUE			08 Alarm Summary	
"DB_Alarms" Alarm.AnyTrip	Bool	TRUE			08 Alarm Summary	
"DB_Alarms" Alarm.FirstOutCode	DEC	1403			08 Alarm Summary	
"PHL_DQ_XV101_OpenCmd"	%Q200.0	Bool	FALSE			
"PHL_DQ_XV102_OpenCmd"	%Q200.1	Bool	FALSE			
"PHL_DQ_XV103_OpenCmd"	%Q200.2	Bool	FALSE			
"PHL_DQ_P102_StartCmd"	%Q200.3	Bool	FALSE			
"DB_HMA" Status.Valve[1].Fault	Bool	FALSE				
"DB_HMA" Status.Valve[1].FaultCode	DEC	0				
"DB_Simulation" Inject.FailValveOpenFb[1]	Bool	FALSE				
"DB_Simulation" Inject.FailValveOpenFb[2]	Bool	FALSE				
"DB_Simulation" Inject.FailValveOpenFb[3]	Bool	TRUE	TRUE			
"DB_HMA" Status.Valve[3].Fault	Bool	TRUE				
"DB_HMA" Status.Valve[3].FaultCode	DEC	1				
"DB_IO" SelectedIn.Valve[3].OpenFb	Bool	FALSE				
"DB_IO" SelectedIn.Pump[2].RunFb	Bool	FALSE				

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Step 90; Valve[3].Fault TRUE / FaultCode 1; AnyAlarm/AnyTrip TRUE; FirstOut 1403.

FAT-17 P101 run-feedback failure

PASS

Check Detect failed-to-start feedback

Observed Pump 1 fault code 1, first-out 1501 and S90 were recorded.

"DB_HMI" Status System SimulationActive	Bool	TRUE	01 PLC Health
"DB_HMI" Status ConfigValid	Bool	TRUE	01 PLC Health
"DB_HMI" Status ConfigProcCode	DEC	0	01 PLC Health
"DB_HMI" Cmd ModeRequest	DEC	2	02 Commands
"DB_HMI" Cmd Start	Bool	FALSE	02 Commands
"DB_HMI" Cmd Stop	Bool	FALSE	02 Commands
"DB_HMI" Cmd RestartPuls	Bool	FALSE	02 Commands
"DB_HMI" Cmd ResetCycle	Bool	FALSE	02 Commands
"DB_HMI" Cmd ResetSimulation	Bool	FALSE	02 Commands
"DB_HMI" Status System Ready	Bool	TRUE	03 System Status
"DB_HMI" Status System ControlEnabled	Bool	TRUE	03 System Status
"DB_HMI" Status System ActiveMode	Bool	FALSE	03 System Status
"DB_HMI" Status System Offline	Bool	FALSE	03 System Status
"DB_HMI" Status System ManualActive	Bool	TRUE	03 System Status
"DB_HMI" Status System Automatic	Bool	TRUE	03 System Status
"DB_HMI" Status System Faulted	Bool	FALSE	03 System Status
"DB_HMI" Status Sequence Stop	Bool	FALSE	04 Sequence
"DB_HMI" Status Sequence Active	Bool	FALSE	04 Sequence
"DB_HMI" Status Sequence Complete	Bool	FALSE	04 Sequence
"DB_HMI" Status Sequence StepTimeout	Bool	FALSE	04 Sequence
"DB_HMI" Status Sequence TimeoutStep	DEC	0	04 Sequence
"DB_HMI" Status Sequence StepRepeatSec	Floatingpoint number	0.019	04 Sequence
"DB_HMI" Status Sequence LastCycleTimeSec	Floatingpoint number	95.1912	04 Sequence
"DB_HMI" Status Sequence CycleCounter	DEC	2	04 Sequence
"DB_HMI" Status Tank1[1]LevelPct	Floatingpoint number	0.0	05 T101 Level
"DB_HMI" Status Tank2[2]LevelPct	Floatingpoint number	0.0	05 T102 Level
"DB_HMI" Status Tank3[3]LevelPct	Floatingpoint number	0.0	05 T103 Level
"DB_OP" ProcessOutValveOpenCmd[1]	Bool	FALSE	06 Process Commands
"DB_OP" ProcessOutValveOpenCmd[2]	Bool	TRUE	06 Process Commands
"DB_OP" ProcessOutValveOpenCmd[3]	Bool	FALSE	06 Process Commands
"DB_OP" ProcessOutValveOpenCmd[4]	Bool	FALSE	06 Process Commands
"DB_OP" ProcessOutValveOpenCmd[5]	Bool	FALSE	06 Process Commands
"DB_OP" ProcessOutValveOpenCmd[6]	Bool	FALSE	06 Process Commands
"DB_OP" PhysicalOutValveOpenCmd[1]	Bool	FALSE	07 Physical Output Gate
"DB_OP" PhysicalOutValveOpenCmd[2]	Bool	FALSE	07 Physical Output Gate
"DB_OP" PhysicalOutValveOpenCmd[3]	Bool	FALSE	07 Physical Output Gate
"DB_OP" PhysicalOutValveOpenCmd[4]	Bool	FALSE	07 Physical Output Gate
"DB_OP" PhysicalOutValveOpenCmd[5]	Bool	FALSE	07 Physical Output Gate
"DB_OP" PhysicalOutValveOpenCmd[6]	Bool	FALSE	07 Physical Output Gate
"DB_Alarm" Alarm Activated	Bool	FALSE	08 Alarm Summary
"DB_Alarm" Alarm AnyAlarm	Bool	FALSE	08 Alarm Summary
"DB_Alarm" Alarm AnyTrip	Bool	FALSE	08 Alarm Summary
"DB_Alarm" Alarm FirstOutCode	DEC	0	08 Alarm Summary
"PLD_Q" J1V101_OpenCmd"	Bool	FALSE	
"PLD_Q" J1V102_OpenCmd"	Bool	FALSE	
"PLD_Q" J1V103_OpenCmd"	Bool	FALSE	
"PLD_Q" J1V104_OpenCmd"	Bool	FALSE	
"DB_HMI" Status Valve1[1]Fault	Bool	FALSE	
"DB_HMI" Status Valve2[2]Fault	Bool	FALSE	
"DB_HMI" Status Valve3[3]Fault	Bool	FALSE	
"DB_HMI" Status Valve4[4]Fault	Bool	FALSE	
"DB_HMI" Status Valve5[5]Fault	Bool	FALSE	
"DB_HMI" Status Valve6[6]Fault	Bool	FALSE	
"DB_OP" SelectInValve[1]OpenPb	Bool	FALSE	
"DB_OP" SelectInValve[2]OpenPb	Bool	FALSE	
"DB_OP" SelectInValve[3]OpenPb	Bool	FALSE	
"DB_OP" SelectInValve[4]OpenPb	Bool	FALSE	
"DB_OP" SelectInValve[5]OpenPb	Bool	FALSE	
"DB_OP" SelectInValve[6]OpenPb	Bool	FALSE	
"DB_HMI" Status Pump1[1]FaultCode	DEC	0	
"DB_HMI" Status Pump2[2]FaultCode	DEC	0	
"DB_HMI" Status Pump3[3]FaultCode	DEC	0	
"DB_HMI" Status Pump4[4]FaultCode	DEC	0	
"DB_HMI" Status Pump5[5]FaultCode	DEC	0	
"DB_HMI" Status Pump6[6]FaultCode	DEC	0	

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P101 RunFb-failure injection active in the transfer sequence; RunFb suppressed.

"DB_HMI" Status System SimulationActive	Bool	TRUE	01 PLC Health
"DB_HMI" Status ConfigValid	Bool	TRUE	01 PLC Health
"DB_HMI" Status ConfigProcCode	DEC	0	01 PLC Health
"DB_HMI" Cmd ModeRequest	DEC	2	02 Commands
"DB_HMI" Cmd Start	Bool	FALSE	02 Commands
"DB_HMI" Cmd Stop	Bool	FALSE	02 Commands
"DB_HMI" Cmd RestartPuls	Bool	FALSE	02 Commands
"DB_HMI" Cmd ResetCycle	Bool	FALSE	02 Commands
"DB_HMI" Cmd ResetSimulation	Bool	FALSE	02 Commands
"DB_HMI" Status System Ready	Bool	TRUE	03 System Status
"DB_HMI" Status System ControlEnabled	Bool	TRUE	03 System Status
"DB_HMI" Status System ActiveMode	Bool	FALSE	03 System Status
"DB_HMI" Status System Offline	Bool	FALSE	03 System Status
"DB_HMI" Status System ManualActive	Bool	TRUE	03 System Status
"DB_HMI" Status System Automatic	Bool	TRUE	03 System Status
"DB_HMI" Status System Faulted	Bool	TRUE	03 System Status
"DB_HMI" Status Sequence Stop	Bool	FALSE	04 Sequence
"DB_HMI" Status Sequence Active	Bool	FALSE	04 Sequence
"DB_HMI" Status Sequence Complete	Bool	FALSE	04 Sequence
"DB_HMI" Status Sequence StepTimeout	Bool	FALSE	04 Sequence
"DB_HMI" Status Sequence TimeoutStep	DEC	0	04 Sequence
"DB_HMI" Status Sequence StepRepeatSec	Floatingpoint number	0.0	04 Sequence
"DB_HMI" Status Sequence LastCycleTimeSec	Floatingpoint number	95.1912	04 Sequence
"DB_HMI" Status Sequence CycleCounter	DEC	2	04 Sequence
"DB_HMI" Status Tank1[1]LevelPct	Floatingpoint number	0.0	05 T101 Level
"DB_HMI" Status Tank2[2]LevelPct	Floatingpoint number	0.0	05 T102 Level
"DB_HMI" Status Tank3[3]LevelPct	Floatingpoint number	0.0	05 T103 Level
"DB_OP" ProcessOutValveOpenCmd[1]	Bool	FALSE	06 Process Commands
"DB_OP" ProcessOutValveOpenCmd[2]	Bool	FALSE	06 Process Commands
"DB_OP" ProcessOutValveOpenCmd[3]	Bool	FALSE	06 Process Commands
"DB_OP" ProcessOutValveOpenCmd[4]	Bool	FALSE	06 Process Commands
"DB_OP" ProcessOutValveOpenCmd[5]	Bool	FALSE	06 Process Commands
"DB_OP" ProcessOutValveOpenCmd[6]	Bool	FALSE	06 Process Commands
"DB_OP" PhysicalOutValveOpenCmd[1]	Bool	FALSE	07 Physical Output Gate
"DB_OP" PhysicalOutValveOpenCmd[2]	Bool	FALSE	07 Physical Output Gate
"DB_OP" PhysicalOutValveOpenCmd[3]	Bool	FALSE	07 Physical Output Gate
"DB_OP" PhysicalOutValveOpenCmd[4]	Bool	FALSE	07 Physical Output Gate
"DB_OP" PhysicalOutValveOpenCmd[5]	Bool	FALSE	07 Physical Output Gate
"DB_OP" PhysicalOutValveOpenCmd[6]	Bool	FALSE	07 Physical Output Gate
"DB_Alarm" Alarm Activated	Bool	FALSE	08 Alarm Summary
"DB_Alarm" Alarm AnyAlarm	Bool	FALSE	08 Alarm Summary
"DB_Alarm" Alarm AnyTrip	Bool	FALSE	08 Alarm Summary
"DB_Alarm" Alarm FirstOutCode	DEC	1501	08 Alarm Summary
"PLD_Q" J1V101_OpenCmd"	Bool	FALSE	
"PLD_Q" J1V102_OpenCmd"	Bool	FALSE	
"PLD_Q" J1V103_OpenCmd"	Bool	FALSE	
"PLD_Q" J1V104_OpenCmd"	Bool	FALSE	
"DB_HMI" Status Valve1[1]Fault	Bool	FALSE	
"DB_HMI" Status Valve2[2]Fault	Bool	FALSE	
"DB_HMI" Status Valve3[3]Fault	Bool	FALSE	
"DB_HMI" Status Valve4[4]Fault	Bool	FALSE	
"DB_HMI" Status Valve5[5]Fault	Bool	FALSE	
"DB_HMI" Status Valve6[6]Fault	Bool	FALSE	
"DB_OP" SelectInValve[1]OpenPb	Bool	FALSE	
"DB_OP" SelectInValve[2]OpenPb	Bool	FALSE	
"DB_OP" SelectInValve[3]OpenPb	Bool	FALSE	
"DB_OP" SelectInValve[4]OpenPb	Bool	FALSE	
"DB_OP" SelectInValve[5]OpenPb	Bool	FALSE	
"DB_OP" SelectInValve[6]OpenPb	Bool	FALSE	
"DB_HMI" Status Pump1[1]FaultCode	DEC	1	
"DB_HMI" Status Pump2[2]FaultCode	DEC	0	
"DB_HMI" Status Pump3[3]FaultCode	DEC	0	
"DB_HMI" Status Pump4[4]FaultCode	DEC	0	
"DB_HMI" Status Pump5[5]FaultCode	DEC	0	
"DB_HMI" Status Pump6[6]FaultCode	DEC	0	

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Step 90; Pump[1].Fault TRUE / FaultCode 1; AnyAlarm/AnyTrip TRUE; FirstOut 1501.

Check Trip on overload during operation

Observed Overload forced Pump 2 fault code 3, first-out 1502 and S90.

"DB_HMI" Status System SimulationActive	Bool	TRUE	01 PLC Health	
"DB_HMI" Status ConfigValid	Bool	TRUE	01 PLC Health	
"DB_HMI" Status ConfigProcCode	DEC	0	01 PLC Health	
"DB_HMI" Cmd ModeRequest	DEC	0	02 Commands	0 OR 1 Manual, 2 Auto
"DB_HMI" Cmd Start	Bool	FALSE	02 Commands	
"DB_HMI" Cmd Stop	Bool	FALSE	02 Commands	
"DB_HMI" Cmd RestartPuls	Bool	FALSE	02 Commands	
"DB_HMI" Cmd RestartCycle	Bool	FALSE	02 Commands	
"DB_HMI" Cmd ResetSimulation	Bool	FALSE	02 Commands	
"DB_HMI" Status System Ready	Bool	TRUE	03 System Status	
"DB_HMI" Status System ControlEnabled	DEC	2	03 System Status	
"DB_HMI" Status System OffActive	Bool	FALSE	03 System Status	0 OR 1 Manual, 2 Auto
"DB_HMI" Status System ManualActive	Bool	TRUE	03 System Status	
"DB_HMI" Status System Automatic	Bool	TRUE	03 System Status	
"DB_HMI" Status System Faulted	Bool	FALSE	03 System Status	
"DB_HMI" Status Sequence Stop	DEC	0	04 Sequence	
"DB_HMI" Status Sequence Active	Bool	FALSE	04 Sequence	
"DB_HMI" Status Sequence Complete	Bool	FALSE	04 Sequence	
"DB_HMI" Status Sequence StepTimeout	Bool	FALSE	04 Sequence	
"DB_HMI" Status Sequence TimeoutStep	DEC	0	04 Sequence	
"DB_HMI" Status Sequence StepRepeatSec	Floatingpoint number	5.338	04 Sequence	
"DB_HMI" Status Sequence LastCycleTimeSec	Floatingpoint number	95.1912	04 Sequence	
"DB_HMI" Status Sequence CycleCounter	DEC	2	04 Sequence	
"DB_HMI" Status Tank1[1]LevelPct	Floatingpoint number	89.0	05 T102 Level	
"DB_HMI" Status Tank2[2]LevelPct	Floatingpoint number	74.0	05 T102 Level	
"DB_HMI" Status Tank3[3]LevelPct	Floatingpoint number	6.0	05 T102 Level	
"DB_OP" ProcessOutValveOpenCmd[1]	Bool	FALSE	06 Process Commands	
"DB_OP" ProcessOutValveOpenCmd[2]	Bool	TRUE	06 Process Commands	
"DB_OP" ProcessOutValveOpenCmd[3]	Bool	TRUE	06 Process Commands	
"DB_OP" ProcessOutValveOpenCmd[4]	Bool	FALSE	06 Process Commands	
"DB_OP" ProcessOutValveOpenCmd[5]	Bool	FALSE	06 Process Commands	
"DB_OP" ProcessOutValveOpenCmd[6]	Bool	FALSE	06 Process Commands	
"DB_OP" PhysicalOutValveOpenCmd[1]	Bool	FALSE	07 Physical Output Gate	
"DB_OP" PhysicalOutValveOpenCmd[2]	Bool	FALSE	07 Physical Output Gate	
"DB_OP" PhysicalOutValveOpenCmd[3]	Bool	FALSE	07 Physical Output Gate	
"DB_OP" PhysicalOutValveOpenCmd[4]	Bool	FALSE	07 Physical Output Gate	
"DB_OP" PhysicalOutValveOpenCmd[5]	Bool	FALSE	07 Physical Output Gate	
"DB_OP" PhysicalOutValveOpenCmd[6]	Bool	FALSE	07 Physical Output Gate	
"DB_Alarm" Alarm Activated	Bool	FALSE	08 Alarm Summary	
"DB_Alarm" Alarm AnyAlarm	Bool	FALSE	08 Alarm Summary	
"DB_Alarm" Alarm AnyFb	Bool	FALSE	08 Alarm Summary	
"DB_Alarm" Alarm FirstOutCode	DEC	0	08 Alarm Summary	
"PLD_Q_V101_OpenCmd"	Bool	FALSE		
"PLD_Q_V102_OpenCmd"	Bool	FALSE		
"PLD_Q_V103_OpenCmd"	Bool	FALSE		
"PLD_Q_V102_StartCmd"	Bool	FALSE		
"DB_HMI" Status Valve1[1]Fault	Bool	FALSE		
"DB_HMI" Status Valve1[1]FaultCode	DEC	0		
"DB_Simulation" InjectFaultValveOpenP[1]	Bool	FALSE		
"DB_Simulation" InjectFaultValveOpenP[2]	Bool	FALSE		
"DB_Simulation" InjectFaultValveOpenP[3]	Bool	FALSE		
"DB_HMI" Status Valve3[3]Fault	Bool	FALSE		
"DB_HMI" Status Valve3[3]FaultCode	DEC	0		
"DB_OP" SelectInPump[2]RunPb	Bool	TRUE		
"DB_Simulation" InjectTripPumpOverload[2]	Bool	FALSE		
"DB_HMI" Status Pump[2]Fault	Bool	FALSE		
"DB_HMI" Status Pump[2]FaultCode	DEC	0		
"DB_OP" SelectInPump[2]OverloadOK	Bool	TRUE		
"DB_OP" SelectInPump[2]RunPb	Bool	TRUE		

"DB_HMI" Status System SimulationActive	Bool	TRUE	01 PLC Health	
"DB_HMI" Status ConfigValid	Bool	TRUE	01 PLC Health	
"DB_HMI" Status ConfigProcCode	DEC	0	01 PLC Health	
"DB_HMI" Cmd ModeRequest	DEC	0	02 Commands	0 OR 1 Manual, 2 Auto
"DB_HMI" Cmd Start	Bool	FALSE	02 Commands	
"DB_HMI" Cmd Stop	Bool	FALSE	02 Commands	
"DB_HMI" Cmd RestartPuls	Bool	FALSE	02 Commands	
"DB_HMI" Cmd RestartCycle	Bool	FALSE	02 Commands	
"DB_HMI" Cmd ResetSimulation	Bool	FALSE	02 Commands	
"DB_HMI" Status System Ready	Bool	FALSE	03 System Status	
"DB_HMI" Status System ControlEnabled	DEC	2	03 System Status	
"DB_HMI" Status System OffActive	Bool	FALSE	03 System Status	0 OR 1 Manual, 2 Auto
"DB_HMI" Status System ManualActive	Bool	FALSE	03 System Status	
"DB_HMI" Status System Automatic	Bool	TRUE	03 System Status	
"DB_HMI" Status System Faulted	Bool	TRUE	03 System Status	
"DB_HMI" Status Sequence Stop	DEC	0	04 Sequence	
"DB_HMI" Status Sequence Active	Bool	FALSE	04 Sequence	
"DB_HMI" Status Sequence Complete	Bool	FALSE	04 Sequence	
"DB_HMI" Status Sequence StepTimeout	Bool	FALSE	04 Sequence	
"DB_HMI" Status Sequence TimeoutStep	DEC	0	04 Sequence	
"DB_HMI" Status Sequence StepRepeatSec	Floatingpoint number	6.0	04 Sequence	
"DB_HMI" Status Sequence LastCycleTimeSec	Floatingpoint number	95.1912	04 Sequence	
"DB_HMI" Status Sequence CycleCounter	DEC	2	04 Sequence	
"DB_HMI" Status Tank1[1]LevelPct	Floatingpoint number	89.0	05 T102 Level	
"DB_HMI" Status Tank2[2]LevelPct	Floatingpoint number	59.0	05 T102 Level	
"DB_HMI" Status Tank3[3]LevelPct	Floatingpoint number	29.0	05 T102 Level	
"DB_OP" ProcessOutValveOpenCmd[1]	Bool	FALSE	06 Process Commands	
"DB_OP" ProcessOutValveOpenCmd[2]	Bool	FALSE	06 Process Commands	
"DB_OP" ProcessOutValveOpenCmd[3]	Bool	FALSE	06 Process Commands	
"DB_OP" ProcessOutValveOpenCmd[4]	Bool	FALSE	06 Process Commands	
"DB_OP" ProcessOutValveOpenCmd[5]	Bool	FALSE	06 Process Commands	
"DB_OP" ProcessOutValveOpenCmd[6]	Bool	FALSE	06 Process Commands	
"DB_OP" PhysicalOutValveOpenCmd[1]	Bool	FALSE	07 Physical Output Gate	
"DB_OP" PhysicalOutValveOpenCmd[2]	Bool	FALSE	07 Physical Output Gate	
"DB_OP" PhysicalOutValveOpenCmd[3]	Bool	FALSE	07 Physical Output Gate	
"DB_OP" PhysicalOutValveOpenCmd[4]	Bool	FALSE	07 Physical Output Gate	
"DB_OP" PhysicalOutValveOpenCmd[5]	Bool	FALSE	07 Physical Output Gate	
"DB_OP" PhysicalOutValveOpenCmd[6]	Bool	FALSE	07 Physical Output Gate	
"DB_Alarm" Alarm Activated	Bool	TRUE	08 Alarm Summary	
"DB_Alarm" Alarm AnyAlarm	Bool	TRUE	08 Alarm Summary	
"DB_Alarm" Alarm AnyFb	Bool	TRUE	08 Alarm Summary	
"DB_Alarm" Alarm FirstOutCode	DEC	1502	08 Alarm Summary	
"PLD_Q_V101_OpenCmd"	Bool	FALSE		
"PLD_Q_V102_OpenCmd"	Bool	FALSE		
"PLD_Q_V103_OpenCmd"	Bool	FALSE		
"PLD_Q_V102_StartCmd"	Bool	FALSE		
"DB_HMI" Status Valve1[1]Fault	Bool	FALSE		
"DB_HMI" Status Valve1[1]FaultCode	DEC	0		
"DB_Simulation" InjectFaultValveOpenP[1]	Bool	FALSE		
"DB_Simulation" InjectFaultValveOpenP[2]	Bool	FALSE		
"DB_Simulation" InjectFaultValveOpenP[3]	Bool	FALSE		
"DB_HMI" Status Valve3[3]Fault	Bool	FALSE		
"DB_HMI" Status Valve3[3]FaultCode	DEC	0		
"DB_OP" SelectInPump[2]RunPb	Bool	TRUE		
"DB_Simulation" InjectTripPumpOverload[2]	Bool	TRUE		
"DB_HMI" Status Pump[2]Fault	Bool	TRUE		
"DB_HMI" Status Pump[2]FaultCode	DEC	3		
"DB_OP" SelectInPump[2]OverloadOK	Bool	FALSE		
"DB_OP" SelectInPump[2]RunPb	Bool	FALSE		

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Before overload: Step 60, P102 command/run feedback TRUE, OverloadOK TRUE, TripPumpOverload[2] FALSE.

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After injection: Step 90; TripPumpOverload[2] TRUE; Pump[2].Fault TRUE / FaultCode 3; OverloadOK FALSE; RunFb FALSE; FirstOut 1502.

Check Remove process demand on simulated safety-chain trip

Observed SafetyCircuitOK went false; the sequence entered S90, alarms/trips latched and first-out 1001 was recorded.

Scope note: standard PLC simulation input used to validate trip response; no certified functional-safety function is claimed.

"DB_HMI" Cmd ModeRequest	DEC	2	FALSE	02 Commands	0 OR 1 Manual, 2 Auto
"DB_HMI" Cmd Start	Bool	0	FALSE	02 Commands	
"DB_HMI" Cmd Stop	Bool	0	FALSE	02 Commands	
"DB_HMI" Cmd ResetFaults	Bool	0	FALSE	02 Commands	
"DB_HMI" Cmd ResetCycle	Bool	0	FALSE	02 Commands	
"DB_HMI" Cmd ResetSimulation	Bool	0	FALSE	02 Commands	
"DB_HMI" Status System Ready	Bool	0	FALSE	03 System Status	
"DB_HMI" Status System ControlEnabled	Bool	0	TRUE	03 System Status	
"DB_HMI" Status System ActivateWkde	DEC	2	FALSE	03 System Status	0 OR 1 Manual, 2 Auto
"DB_HMI" Status System DeactivateWkde	Bool	0	FALSE	03 System Status	
"DB_HMI" Status System ManualActive	Bool	0	FALSE	03 System Status	
"DB_HMI" Status System Automatic	Bool	0	TRUE	03 System Status	
"DB_HMI" Status System Faulted	Bool	0	FALSE	03 System Status	
"DB_HMI" Status Sequence Active	Bool	0	FALSE	04 Sequence	
"DB_HMI" Status Sequence Complete	DEC	40	FALSE	04 Sequence	
"DB_HMI" Status Sequence Step	DEC	0	FALSE	04 Sequence	
"DB_HMI" Status Sequence StepTimeOut	DEC	0	FALSE	04 Sequence	
"DB_HMI" Status Sequence TimeOutStep	DEC	0	FALSE	04 Sequence	
"DB_HMI" Status Sequence StepExpandedSec	Flashingpoint number	5.714	FALSE	04 Sequence	
"DB_HMI" Status Sequence LastCycleTimeOutSec	Flashingpoint number	18.1912	FALSE	04 Sequence	
"DB_HMI" Status Sequence CycleCounter	DEC	2	FALSE	04 Sequence	
"DB_HMI" Status Tank1LevelPct	Flashingpoint number	72.0	FALSE	05 T101 Level	
"DB_HMI" Status Tank2LevelPct	Flashingpoint number	8.0	FALSE	05 T102 Level	
"DB_HMI" Status Tank3LevelPct	Flashingpoint number	0.0	FALSE	05 T103 Level	
"DB_OP" ProcessOutValveOpenCmd[1]	Bool	0	TRUE	06 Process Commands	
"DB_OP" ProcessOutValveOpenCmd[2]	Bool	0	TRUE	06 Process Commands	
"DB_OP" ProcessOutValveOpenCmd[3]	Bool	0	FALSE	06 Process Commands	
"DB_OP" ProcessOutPumpStartCmd[1]	Bool	0	TRUE	06 Process Commands	
"DB_OP" ProcessOutPumpStartCmd[2]	Bool	0	FALSE	06 Process Commands	
"DB_OP" ProcessOutPumpStartCmd[3]	Bool	0	FALSE	06 Process Commands	
"DB_OP" PhysicalOutValveOpenCmd[1]	Bool	0	FALSE	07 Physical Output Gate	
"DB_OP" PhysicalOutValveOpenCmd[2]	Bool	0	FALSE	07 Physical Output Gate	
"DB_OP" PhysicalOutValveOpenCmd[3]	Bool	0	FALSE	07 Physical Output Gate	
"DB_OP" PhysicalOutPumpStartCmd[1]	Bool	0	FALSE	07 Physical Output Gate	
"DB_OP" PhysicalOutPumpStartCmd[2]	Bool	0	FALSE	07 Physical Output Gate	
"DB_OP" PhysicalOutPumpStartCmd[3]	Bool	0	FALSE	07 Physical Output Gate	
"DB_Alarm" Alarm Activated	Bool	1600000_2000	TRUE	08 Alarm Summary	
"DB_Alarm" Alarm AnyAlarm	Bool	0	FALSE	08 Alarm Summary	
"DB_Alarm" Alarm AnyTrip	Bool	0	FALSE	08 Alarm Summary	
"DB_Alarm" Alarm FirstOutCode	DEC	0	FALSE	08 Alarm Summary	
"PLD_Q_V101_OpenCmd"	Bool	1600000_2000	FALSE	08 Alarm Summary	
"PLD_Q_V102_OpenCmd"	Bool	1600000_2000	FALSE	08 Alarm Summary	
"PLD_Q_V103_OpenCmd"	Bool	1600000_2000	FALSE	08 Alarm Summary	
"PLD_Q_V104_OpenCmd"	Bool	1600000_2000	FALSE	08 Alarm Summary	
"DB_HMI" Status Valve1Fault	DEC	0	FALSE	08 Alarm Summary	
"DB_Simulation" InjectFaultValveOpenP[1]	Bool	0	FALSE	08 Alarm Summary	
"DB_Simulation" InjectFaultValveOpenP[2]	Bool	0	FALSE	08 Alarm Summary	
"DB_Simulation" InjectFaultValveOpenP[3]	Bool	0	FALSE	08 Alarm Summary	
"DB_HMI" Status Valve1FaultCode	DEC	0	FALSE	08 Alarm Summary	
"DB_HMI" Status Valve2FaultCode	DEC	0	FALSE	08 Alarm Summary	
"DB_HMI" Status Valve3FaultCode	DEC	0	FALSE	08 Alarm Summary	
"DB_OP" SelectedInValve[1]OpenP	Bool	0	FALSE	08 Alarm Summary	
"DB_OP" SelectedInValve[2]OpenP	Bool	0	FALSE	08 Alarm Summary	
"DB_Simulation" InjectTripPumpOverhaul[2]	Bool	0	FALSE	08 Alarm Summary	
"DB_HMI" Status Pump2Fault	DEC	0	FALSE	08 Alarm Summary	
"DB_HMI" Status Pump3FaultCode	DEC	0	FALSE	08 Alarm Summary	
"DB_OP" SelectedInPump[2]Overhaul	Bool	0	FALSE	08 Alarm Summary	
"DB_OP" SelectedInPump[2]Overhaul	Bool	0	FALSE	08 Alarm Summary	
"DB_OP" SelectedInPump[2]Overhaul	Bool	0	FALSE	08 Alarm Summary	
"DB_Simulation" InjectTripSafetyCircuit	Bool	0	FALSE	08 Alarm Summary	
"DB_OP" SelectedInSafetyCircuit	Bool	0	FALSE	08 Alarm Summary	

TT-F19-A

Pre-trip S40: SafetyCircuitOK TRUE, no alarm/trip, active logical process requests.

"DB_HMI" Cmd ModeRequest	DEC	2	FALSE	02 Commands	0 OR 1 Manual, 2 Auto
"DB_HMI" Cmd Start	Bool	0	FALSE	02 Commands	
"DB_HMI" Cmd Stop	Bool	0	FALSE	02 Commands	
"DB_HMI" Cmd ResetFaults	Bool	0	FALSE	02 Commands	
"DB_HMI" Cmd ResetCycle	Bool	0	FALSE	02 Commands	
"DB_HMI" Cmd ResetSimulation	Bool	0	FALSE	02 Commands	
"DB_HMI" Status System Ready	Bool	0	FALSE	03 System Status	
"DB_HMI" Status System ControlEnabled	Bool	0	FALSE	03 System Status	
"DB_HMI" Status System ActivateWkde	DEC	2	FALSE	03 System Status	0 OR 1 Manual, 2 Auto
"DB_HMI" Status System DeactivateWkde	Bool	0	FALSE	03 System Status	
"DB_HMI" Status System ManualActive	Bool	0	FALSE	03 System Status	
"DB_HMI" Status System Automatic	Bool	0	TRUE	03 System Status	
"DB_HMI" Status System Faulted	Bool	0	FALSE	03 System Status	
"DB_HMI" Status Sequence Active	Bool	0	FALSE	04 Sequence	
"DB_HMI" Status Sequence Complete	DEC	40	FALSE	04 Sequence	
"DB_HMI" Status Sequence Step	DEC	0	FALSE	04 Sequence	
"DB_HMI" Status Sequence StepTimeOut	DEC	0	FALSE	04 Sequence	
"DB_HMI" Status Sequence TimeOutStep	DEC	0	FALSE	04 Sequence	
"DB_HMI" Status Sequence StepExpandedSec	Flashingpoint number	5.714	FALSE	04 Sequence	
"DB_HMI" Status Sequence LastCycleTimeOutSec	Flashingpoint number	18.1912	FALSE	04 Sequence	
"DB_HMI" Status Sequence CycleCounter	DEC	2	FALSE	04 Sequence	
"DB_HMI" Status Tank1LevelPct	Flashingpoint number	72.0	FALSE	05 T101 Level	
"DB_HMI" Status Tank2LevelPct	Flashingpoint number	8.0	FALSE	05 T102 Level	
"DB_HMI" Status Tank3LevelPct	Flashingpoint number	0.0	FALSE	05 T103 Level	
"DB_OP" ProcessOutValveOpenCmd[1]	Bool	0	FALSE	06 Process Commands	
"DB_OP" ProcessOutValveOpenCmd[2]	Bool	0	FALSE	06 Process Commands	
"DB_OP" ProcessOutValveOpenCmd[3]	Bool	0	FALSE	06 Process Commands	
"DB_OP" ProcessOutPumpStartCmd[1]	Bool	0	FALSE	06 Process Commands	
"DB_OP" ProcessOutPumpStartCmd[2]	Bool	0	FALSE	06 Process Commands	
"DB_OP" ProcessOutPumpStartCmd[3]	Bool	0	FALSE	06 Process Commands	
"DB_OP" PhysicalOutValveOpenCmd[1]	Bool	0	FALSE	07 Physical Output Gate	
"DB_OP" PhysicalOutValveOpenCmd[2]	Bool	0	FALSE	07 Physical Output Gate	
"DB_OP" PhysicalOutValveOpenCmd[3]	Bool	0	FALSE	07 Physical Output Gate	
"DB_OP" PhysicalOutPumpStartCmd[1]	Bool	0	FALSE	07 Physical Output Gate	
"DB_OP" PhysicalOutPumpStartCmd[2]	Bool	0	FALSE	07 Physical Output Gate	
"DB_OP" PhysicalOutPumpStartCmd[3]	Bool	0	FALSE	07 Physical Output Gate	
"DB_Alarm" Alarm Activated	Bool	1600000_2000	TRUE	08 Alarm Summary	
"DB_Alarm" Alarm AnyAlarm	Bool	0	FALSE	08 Alarm Summary	
"DB_Alarm" Alarm AnyTrip	Bool	0	FALSE	08 Alarm Summary	
"DB_Alarm" Alarm FirstOutCode	DEC	1001	FALSE	08 Alarm Summary	
"PLD_Q_V101_OpenCmd"	Bool	1600000_2000	FALSE	08 Alarm Summary	
"PLD_Q_V102_OpenCmd"	Bool	1600000_2000	FALSE	08 Alarm Summary	
"PLD_Q_V103_OpenCmd"	Bool	1600000_2000	FALSE	08 Alarm Summary	
"PLD_Q_V104_OpenCmd"	Bool	1600000_2000	FALSE	08 Alarm Summary	
"DB_HMI" Status Valve1Fault	DEC	0	FALSE	08 Alarm Summary	
"DB_Simulation" InjectFaultValveOpenP[1]	Bool	0	FALSE	08 Alarm Summary	
"DB_Simulation" InjectFaultValveOpenP[2]	Bool	0	FALSE	08 Alarm Summary	
"DB_Simulation" InjectFaultValveOpenP[3]	Bool	0	FALSE	08 Alarm Summary	
"DB_HMI" Status Valve1FaultCode	DEC	0	FALSE	08 Alarm Summary	
"DB_HMI" Status Valve2FaultCode	DEC	0	FALSE	08 Alarm Summary	
"DB_HMI" Status Valve3FaultCode	DEC	0	FALSE	08 Alarm Summary	
"DB_OP" SelectedInValve[1]OpenP	Bool	0	FALSE	08 Alarm Summary	
"DB_OP" SelectedInValve[2]OpenP	Bool	0	FALSE	08 Alarm Summary	
"DB_Simulation" InjectTripPumpOverhaul[2]	Bool	0	FALSE	08 Alarm Summary	
"DB_HMI" Status Pump2Fault	DEC	0	FALSE	08 Alarm Summary	
"DB_HMI" Status Pump3FaultCode	DEC	0	FALSE	08 Alarm Summary	
"DB_OP" SelectedInPump[2]Overhaul	Bool	0	FALSE	08 Alarm Summary	
"DB_OP" SelectedInPump[2]Overhaul	Bool	0	FALSE	08 Alarm Summary	
"DB_OP" SelectedInPump[2]Overhaul	Bool	0	FALSE	08 Alarm Summary	
"DB_Simulation" InjectTripSafetyCircuit	Bool	0	FALSE	08 Alarm Summary	
"DB_OP" SelectedInSafetyCircuit	Bool	0	FALSE	08 Alarm Summary	

TT-F19-B

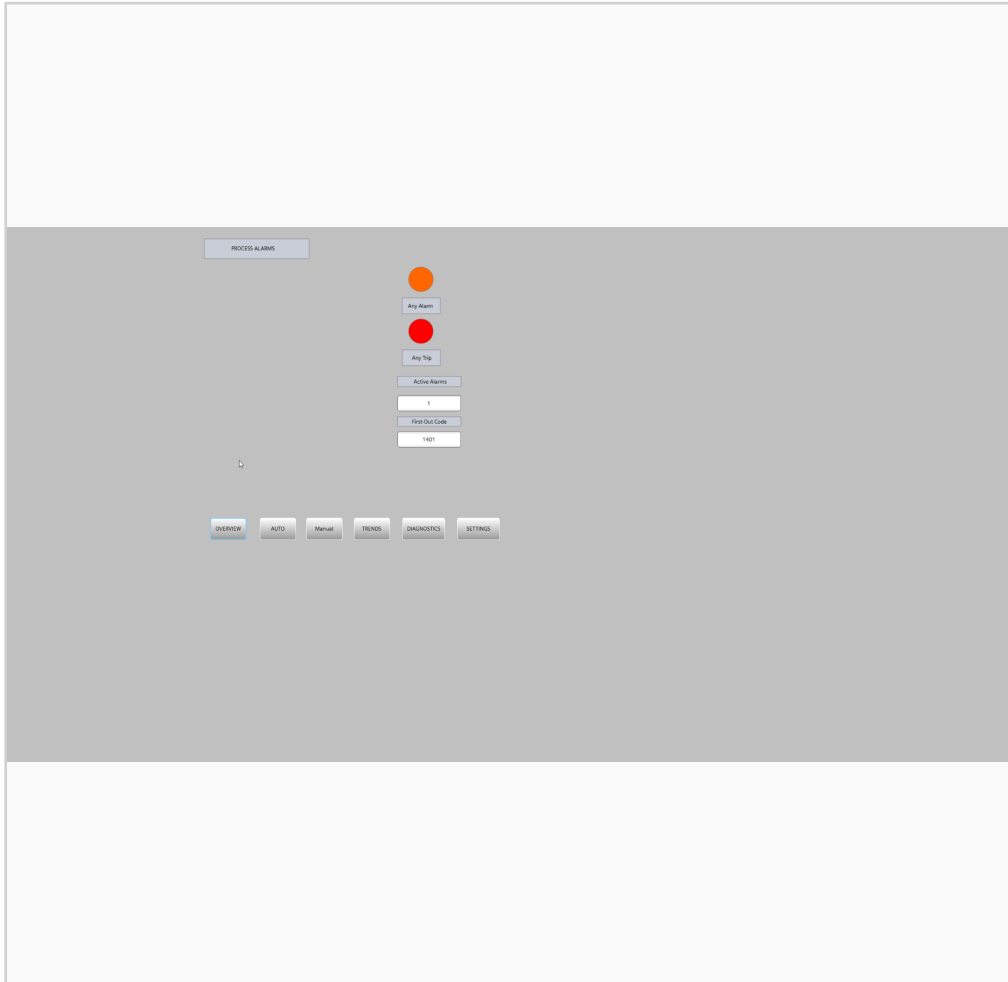
TripSafetyCircuit TRUE, SafetyCircuitOK FALSE, Step 90, Faulted/AnyAlarm/AnyTrip TRUE, FirstOutCode 1001, process requests removed.

FAT-24 Reset and persistent field fault

VERIFIED

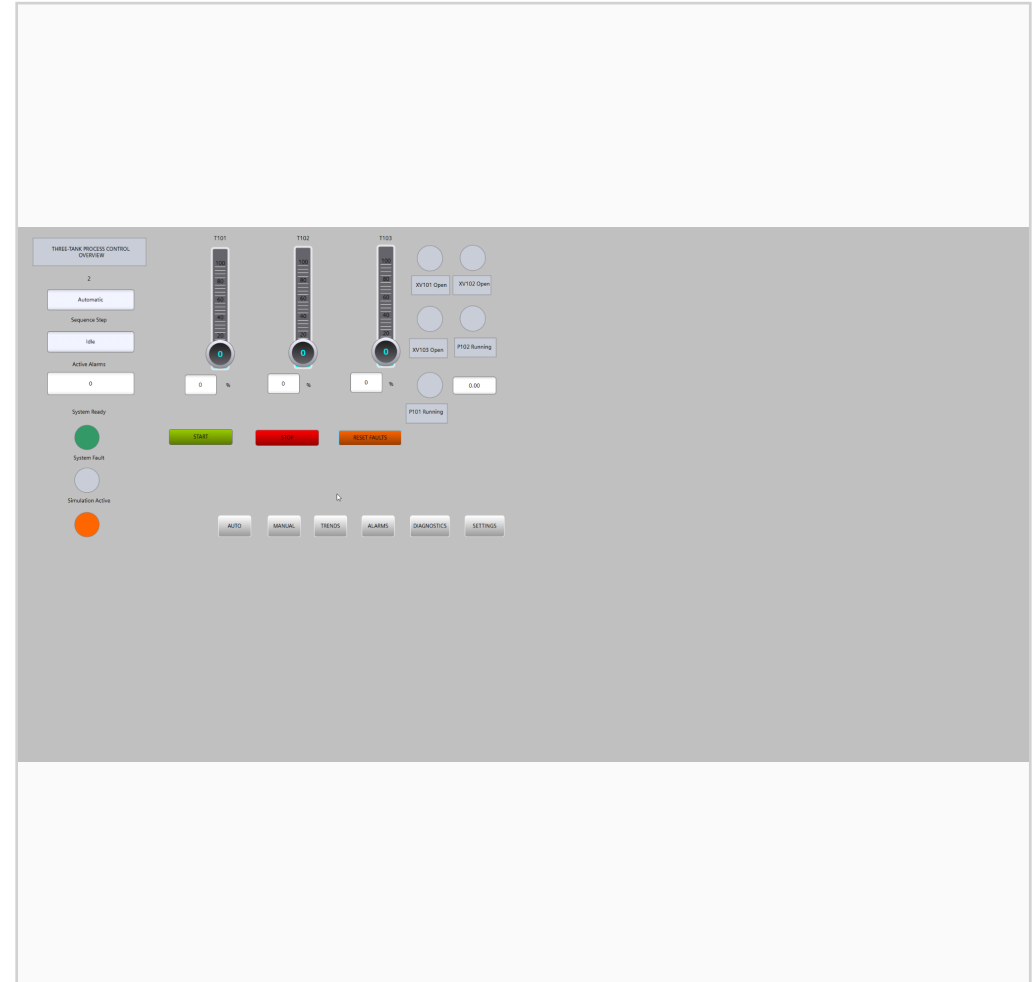
Check Verify controlled reset / deliberate restart behaviour

Observed Reset returned the controller to a stable disabled state without automatic restart. With the injected field fault still present, the fault returned when that device was demanded again.



TT-RESET-A

Fault latched on the HMI with First-Out 1401.



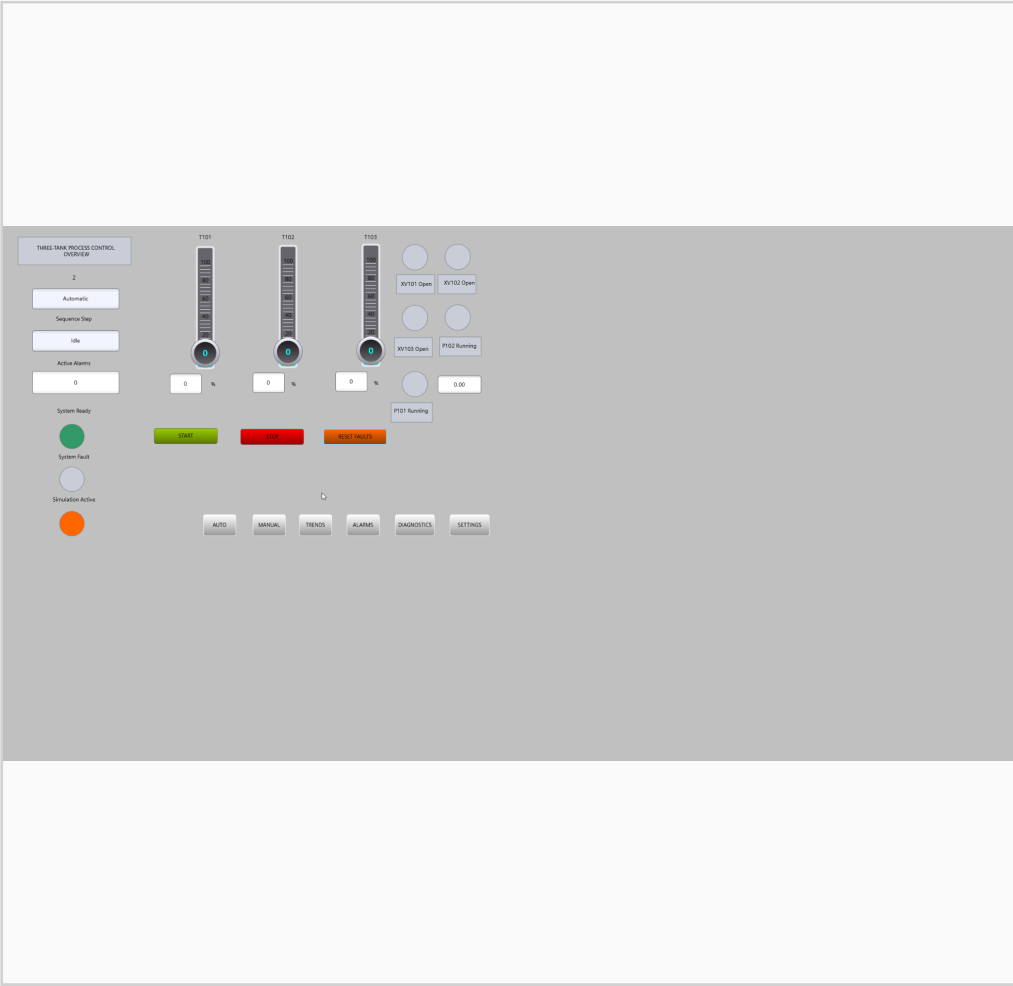
TT-RESET-B

Reset/recovery state captured during the controlled-restart test.

- Check

Recover after the fault cause is cleared
- Observed

Cause cleared -> reset -> deliberate Start restored normal operation.



TT-RESET-B
Recovery sequence after reset.

"DB_System".ScanCounter	DEC	5356936			01 PLC Health		
"DB_Simulation".Status.Tick100ms	Bool	TRUE			01 PLC Health		
"DB_Config".Active.SimulationEnable	Bool	TRUE			01 PLC Health		
"DB_HMA".Status.System.SimulationActive	Bool	TRUE			01 PLC Health		
"DB_HMA".Status.ConfigValid	Bool	TRUE			01 PLC Health		
"DB_HMA".Status.ConfigErrorCode	DEC	0			01 PLC Health		
"DB_HMA".Cmd.ModeRequest	DEC	2			02 Commands	0 Off, 1 Manual, 2 Auto	
"DB_HMA".Cmd.Start	Bool	FALSE	FALSE		02 Commands		
"DB_HMA".Cmd.Stop	Bool	FALSE			02 Commands		
"DB_HMA".Cmd.ResetFaults	Bool	FALSE			02 Commands		
"DB_HMA".Cmd.ResetCycle	Bool	FALSE			02 Commands		
"DB_HMA".Cmd.ResetSimulation	Bool	FALSE			02 Commands		
"DB_HMA".Status.System.Ready	Bool	TRUE			03 System Status		
"DB_HMA".Status.System.ControlEnabled	Bool	TRUE			03 System Status		
"DB_HMA".Status.System.ActiveMode	DEC	2			03 System Status	0 Off, 1 Manual, 2 Auto	
"DB_HMA".Status.System.OffActive	Bool	FALSE			03 System Status		
"DB_HMA".Status.System.ManualActive	Bool	FALSE			03 System Status		
"DB_HMA".Status.System.AutoActive	Bool	TRUE			03 System Status		
"DB_HMA".Status.System.Faulted	Bool	FALSE			03 System Status		
"DB_HMA".Status.Sequence.Step	DEC	20			04 Sequence		
"DB_HMA".Status.Sequence.Active	Bool	FALSE			04 Sequence		
"DB_HMA".Status.Sequence.Complete	Bool	FALSE			04 Sequence		
"DB_HMA".Status.Sequence.StepTimeout	Bool	FALSE			04 Sequence		
"DB_HMA".Status.Sequence.TimeoutStep	DEC	0			04 Sequence		
"DB_HMA".Status.Sequence.StepElapsedSec	Floating-point number	0.854			04 Sequence		
"DB_HMA".Status.Sequence.LastCycleTimeSec	Floating-point number	95.19912			04 Sequence		
"DB_HMA".Status.Sequence.CycleCounter	DEC	2			04 Sequence		
"DB_HMA".Status.Tank[1].LevelPct	Floating-point number	0.0			05 T101 Level		
"DB_HMA".Status.Tank[2].LevelPct	Floating-point number	0.0			05 T102 Level		
"DB_HMA".Status.Tank[3].LevelPct	Floating-point number	0.0			05 T103 Level		
"DB_IO".ProcessOut.ValveOpenCmd[1]	Bool	TRUE			06 Process Commands		
"DB_IO".ProcessOut.ValveOpenCmd[2]	Bool	FALSE			06 Process Commands		
"DB_IO".ProcessOut.ValveOpenCmd[3]	Bool	FALSE			06 Process Commands		
"DB_IO".ProcessOut.PumpStartCmd[1]	Bool	FALSE			06 Process Commands		
"DB_IO".ProcessOut.PumpStartCmd[2]	Bool	FALSE			06 Process Commands		
"DB_IO".PhysicalOut.ValveOpenCmd[1]	Bool	FALSE			07 Physical Output Gate		
"DB_IO".PhysicalOut.ValveOpenCmd[2]	Bool	FALSE			07 Physical Output Gate		
"DB_IO".PhysicalOut.ValveOpenCmd[3]	Bool	FALSE			07 Physical Output Gate		
"DB_IO".PhysicalOut.PumpStartCmd[1]	Bool	FALSE			07 Physical Output Gate		
"DB_IO".PhysicalOut.PumpStartCmd[2]	Bool	FALSE			07 Physical Output Gate		
"DB_Alarms".Alarm.ActiveWord	Hex	1680000_0000			08 Alarm Summary		
"DB_Alarms".Alarm.AnyAlarm	Bool	FALSE			08 Alarm Summary		
"DB_Alarms".Alarm.AnyTrip	Bool	FALSE			08 Alarm Summary		
"DB_Alarms".Alarm.FirstOutCode	DEC	0			08 Alarm Summary		
"PIL_DQ_XV101_OpenCmd"	%Q200.0	Bool	FALSE				
"PIL_DQ_XV102_OpenCmd"	%Q200.1	Bool	FALSE				
"PIL_DQ_XV103_OpenCmd"	%Q200.2	Bool	FALSE				
"PIL_DQ_P102_StartCmd"	%Q200.3	Bool	FALSE				
"DB_HMA".Status.Valve[1].Fault	Bool	FALSE					
"DB_HMA".Status.Valve[1].FaultCode	DEC	0					
"DB_Simulation".Inject.FaultValveOpenPh[1]	Bool	FALSE					

TT-RESET-C
Normal operation restored after cause clear and deliberate restart.

Check Verify manual valve command path

Observed Manual mode was active and XV101 ProcessOut command was captured true with no alarm/trip.

"DB_HHM".Status.ConfigValid	Bool	☒	TRUE		☐	01 PLC Health	
"DB_HHM".Status.ConfigErrorCode	DEC	☐	0		☐	01 PLC Health	
"DB_HHM".Cmd.ModeRequest	DEC	☐	1		☐	02 Commands	0 Off, 1 Manual, 2 Auto
"DB_HHM".Cmd.Start	Bool	☐	FALSE		☐	02 Commands	
"DB_HHM".Cmd.Stop	Bool	☐	FALSE		☐	02 Commands	
"DB_HHM".Cmd.ResetFaults	Bool	☐	FALSE		☐	02 Commands	
"DB_HHM".Cmd.ResetCycle	Bool	☐	FALSE		☐	02 Commands	
"DB_HHM".Cmd.ResetSimulation	Bool	☒	TRUE		☐	02 Commands	
"DB_HHM".Status.SystemReady	Bool	☒	TRUE		☐	03 System Status	
"DB_HHM".Status.System.ControlEnabled	Bool	☒	TRUE		☐	03 System Status	
"DB_HHM".Status.System.ActiveMode	DEC	☐	1		☐	03 System Status	0 Off, 1 Manual, 2 Auto
"DB_HHM".Status.System.OffActive	Bool	☐	FALSE		☐	03 System Status	
"DB_HHM".Status.System.ManualActive	Bool	☒	TRUE		☐	03 System Status	
"DB_HHM".Status.System.AutoActive	Bool	☐	FALSE		☐	03 System Status	
"DB_HHM".Status.System.Faulted	Bool	☐	FALSE		☐	03 System Status	
"DB_HHM".Status.Sequence.Step	DEC	☐	0		☐	04 Sequence	
"DB_HHM".Status.Sequence.Active	Bool	☐	FALSE		☐	04 Sequence	
"DB_HHM".Status.Sequence.Complete	Bool	☐	FALSE		☐	04 Sequence	
"DB_HHM".Status.Sequence.StepTimeout	Bool	☐	FALSE		☐	04 Sequence	
"DB_HHM".Status.Sequence.TimeoutStep	DEC	☐	0		☐	04 Sequence	
"DB_HHM".Status.Sequence.StepElapsedSec	Floating-point number	☐	0.0		☐	04 Sequence	
"DB_HHM".Status.Sequence.LastCycleTimeSec	Floating-point number	☐	95.19912		☐	04 Sequence	
"DB_HHM".Status.Sequence.CycleCounter	DEC	☐	2		☐	04 Sequence	
"DB_HHM".Status.Tank[1].LevelPct	Floating-point number	☐	36.0		☐	05 TI01 Level	
"DB_HHM".Status.Tank[2].LevelPct	Floating-point number	☐	0.0		☐	05 TI02 Level	
"DB_HHM".Status.Tank[3].LevelPct	Floating-point number	☐	0.0		☐	05 TI03 Level	
"DB_IO".ProcessOut.ValveOpenCmd[1]	Bool	☒	TRUE		☐	06 Process Commands	
"DB_IO".ProcessOut.ValveOpenCmd[2]	Bool	☐	FALSE		☐	06 Process Commands	
"DB_IO".ProcessOut.ValveOpenCmd[3]	Bool	☐	FALSE		☐	06 Process Commands	
"DB_IO".ProcessOut.PumpStartCmd[1]	Bool	☐	FALSE		☐	06 Process Commands	
"DB_IO".ProcessOut.PumpStartCmd[2]	Bool	☐	FALSE		☐	06 Process Commands	
"DB_IO".PhysicalOut.ValveOpenCmd[1]	Bool	☐	FALSE		☐	07 Physical Output Gate	
"DB_IO".PhysicalOut.ValveOpenCmd[2]	Bool	☐	FALSE		☐	07 Physical Output Gate	
"DB_IO".PhysicalOut.ValveOpenCmd[3]	Bool	☐	FALSE		☐	07 Physical Output Gate	
"DB_IO".PhysicalOut.PumpStartCmd[1]	Bool	☐	FALSE		☐	07 Physical Output Gate	
"DB_IO".PhysicalOut.PumpStartCmd[2]	Bool	☐	FALSE		☐	07 Physical Output Gate	
"DB_Alarms".Alarm.ActiveWord	Hex	☐	16#0000_0000		☐	08 Alarm Summary	
"DB_Alarms".Alarm.AnyAlarm	Bool	☐	FALSE		☐	08 Alarm Summary	
"DB_Alarms".Alarm.AnyTrip	Bool	☐	FALSE		☐	08 Alarm Summary	
"DB_Alarms".Alarm.FirstOutCode	DEC	☐	0		☐	08 Alarm Summary	
"PN_DQ_XV101_OpenCmd"	%Q200.0	Bool	☐	FALSE	☐		
"PN_DQ_XV102_OpenCmd"	%Q200.1	Bool	☐	FALSE	☐		
"PN_DQ_XV103_OpenCmd"	%Q200.2	Bool	☐	FALSE	☐		
"PN_DQ_P102_StartCmd"	%Q200.3	Bool	☐	FALSE	☐		
"DB_HHM".Status.Valve[1].Fault	Bool	☐	FALSE		☐		
"DB_HHM".Status.Valve[1].FaultCode	DEC	☐	0		☐		
"DB_Simulation".Inject.FailValveOpenFb[1]	Bool	☐	FALSE		☐		
"DB_Simulation".Inject.FailValveOpenFb[2]	Bool	☐	FALSE		☐		
"DB_Simulation".Inject.FailValveOpenFb[3]	Bool	☐	FALSE		☐		
"DB_HHM".Status.Valve[3].Fault	Bool	☐	FALSE		☐		
"DB_HHM".Status.Valve[3].FaultCode	DEC	☐	0		☐		
"DB_IO".SelectedIn.Valve[3].OpenFb	Bool	☐	FALSE		☐		
"DB_IO".SelectedIn.Pump[2].RunFb	Bool	☐	FALSE		☐		
"DB_Simulation".Inject.TripPumpOverload[2]	Bool	☐	FALSE		☐		
"DB_HHM".Status.Pump[2].Fault	Bool	☐	FALSE		☐		
"DB_HHM".Status.Pump[2].FaultCode	DEC	☐	0		☐		
"DB_IO".SelectedIn.Pump[2].OverloadOK	Bool	☒	TRUE		☐		
"DB_IO".SelectedIn.Pump[2].RunFb	Bool	☐	FALSE		☐		
"DB_Simulation".Inject.TripSafetyCircuit	Bool	☐	FALSE		☐		
"DB_IO".SelectedIn.SafetyCircuitOK	Bool	☒	TRUE		☐		

TT-F26-A

Manual mode with XV101 logical command active.

FAT-27 Manual pump interlock

PASS

Check Verify the manual P101 path remains interlocked until XV102 is proved open.

Observed With XV102 feedback absent, P101 remained inhibited. After XV102 feedback was established, the HMI P101 Permitted indication became active.

Variable Name	Variable Type	Variable Value	Variable Unit	Variable Description	Variable Comment
"DB_HMNT" Status ConfigValid	Bool	TRUE		01 PLC Health	
"DB_HMNT" Status ConfigModeCode	DEC	0		01 PLC Health	
"DB_HMNT" Cmd ModeRequest	DEC	1		02 Commands	0 DB, 1 Manual, 2 Auto
"DB_HMNT" Cmd Start	Bool	FALSE		02 Commands	
"DB_HMNT" Cmd Stop	Bool	FALSE		02 Commands	
"DB_HMNT" Cmd ResetFaults	Bool	FALSE		02 Commands	
"DB_HMNT" Cmd ResetCycle	Bool	FALSE		02 Commands	
"DB_HMNT" Cmd ResetSimulation	Bool	TRUE		02 Commands	
"DB_HMNT" Status System Ready	Bool	TRUE		03 System Status	
"DB_HMNT" Status System ControlEnabled	Bool	TRUE		03 System Status	
"DB_HMNT" Status System ActivationModeCode	DEC	1		03 System Status	0 DB, 1 Manual, 2 Auto
"DB_HMNT" Status System Obsolete	Bool	FALSE		03 System Status	
"DB_HMNT" Status System ManualActive	Bool	TRUE		03 System Status	
"DB_HMNT" Status System Automatic	Bool	FALSE		03 System Status	
"DB_HMNT" Status System Faulted	Bool	FALSE		03 System Status	
"DB_HMNT" Status Sequence Step	DEC	0		04 Sequence	
"DB_HMNT" Status Sequence Active	Bool	FALSE		04 Sequence	
"DB_HMNT" Status Sequence Complete	Bool	FALSE		04 Sequence	
"DB_HMNT" Status Sequence StepTimeout	Bool	FALSE		04 Sequence	
"DB_HMNT" Status Sequence TimeoutStep	DEC	0		04 Sequence	
"DB_HMNT" Status Sequence StepElapseSec	Floating-point number	0.0		04 Sequence	
"DB_HMNT" Status Sequence LastCycleTimeSec	Floating-point number	95.19912		04 Sequence	
"DB_HMNT" Status Sequence CycleCounter	DEC	3		04 Sequence	
"DB_HMNT" Status Tank[1] LevelPct	Floating-point number	41.0		05 T101 Level	
"DB_HMNT" Status Tank[2] LevelPct	Floating-point number	0.0		05 T102 Level	
"DB_HMNT" Status Tank[3] LevelPct	Floating-point number	0.0		05 T103 Level	
"DB_IO" ProcessOutValveOpenCmd[1]	Bool	FALSE		06 Process Commands	
"DB_IO" ProcessOutValveOpenCmd[2]	Bool	TRUE		06 Process Commands	
"DB_IO" ProcessOutValveOpenCmd[3]	Bool	FALSE		06 Process Commands	
"DB_IO" ProcessOutPumpStartCmd[1]	Bool	FALSE		06 Process Commands	
"DB_IO" ProcessOutPumpStartCmd[2]	Bool	FALSE		06 Process Commands	
"DB_IO" PhysicalOutValveOpenCmd[1]	Bool	FALSE		07 Physical Output Gate	
"DB_IO" PhysicalOutValveOpenCmd[2]	Bool	FALSE		07 Physical Output Gate	
"DB_IO" PhysicalOutValveOpenCmd[3]	Bool	FALSE		07 Physical Output Gate	
"DB_IO" PhysicalOutPumpStartCmd[1]	Bool	FALSE		07 Physical Output Gate	
"DB_IO" PhysicalOutPumpStartCmd[2]	Bool	FALSE		07 Physical Output Gate	
"DB_Alarm" Alarm Activated	Hex	16A0000_0000		08 Alarm Summary	
"DB_Alarm" Alarm AnyAlarm	Bool	FALSE		08 Alarm Summary	
"DB_Alarm" Alarm AnyTrip	Bool	FALSE		08 Alarm Summary	
"DB_Alarm" Alarm FirstOccCode	DEC	0		08 Alarm Summary	
"PHL_DQ_XV101_OpenCmd"	Bool	FALSE		08 Alarm Summary	
"PHL_DQ_XV102_OpenCmd"	Bool	FALSE			
"PHL_DQ_XV103_OpenCmd"	Bool	FALSE			
"PHL_DQ_F102_StartCmd"	Bool	FALSE			
"DB_HMNT" Status Valve[1] Fault	Bool	FALSE			
"DB_HMNT" Status Valve[1] FaultCode	DEC	0			
"DB_Simulation" Inject FailActiveOpenB[1]	Bool	FALSE			
"DB_Simulation" Inject FailActiveOpenB[2]	Bool	FALSE			
"DB_Simulation" Inject FailActiveOpenB[3]	Bool	FALSE			
"DB_HMNT" Status Valve[5] Fault	Bool	FALSE			
"DB_HMNT" Status Valve[3] FaultCode	DEC	0			
"DB_IO" SelectedInValve[3] OpenB	Bool	FALSE			
"DB_IO" SelectedInPump[2] RunB	Bool	FALSE			
"DB_Simulation" Inject TripPumpOverload[2]	Bool	FALSE			
"DB_HMNT" Status Pump[2] Fault	Bool	FALSE			
"DB_HMNT" Status Pump[2] FaultCode	DEC	0			
"DB_IO" SelectedInPump[2] OverloadOK	Bool	TRUE			
"DB_IO" SelectedInPump[2] RunB	Bool	FALSE			
"DB_Simulation" Inject TripSafetyCircuit	Bool	FALSE			
"DB_IO" SelectedInSafetyCircuitOK	Bool	TRUE			

Before path proof

Manual mode active; XV102 requested while P101 remains inhibited.

The screenshot displays the 'MANUAL OPERATION' interface. On the left, there are four buttons: 'OFF Mode', 'Manual Mode', 'Manual' (a dropdown menu), and 'RESET FAULTS'. The main area contains a table with five columns: 'Device', 'Manual Command', 'Perm200', 'FeedBack', and 'Feed'. The table has three rows for devices XV101, XV102, and XV103. XV101 and XV102 have 'OPEN' commands, while XV103 has a 'RUN' command. Status indicators (green circles) are present for XV101 and XV102. At the bottom, a navigation bar contains buttons for 'OVERVIEW', 'AUTO', 'TRENDS', 'ALARMS', 'DIAGNOSTICS', and 'SETTINGS'.

Device	Manual Command	Perm200	FeedBack	Feed
XV101	OPEN	●	○	○
XV102	OPEN	●	●	○
XV103	OPEN	○	○	○
P101	RUN	●	○	○
P102	RUN	○	○	○

OVERVIEW AUTO TRENDS ALARMS DIAGNOSTICS SETTINGS

After XV102 proof

XV102 feedback is present and P101 Permitted is active on the Manual page.

Check Verify simulation reset does not create an uncontrolled sequence reset

Observed During an active cycle the simulation-reset request did not create a fault or uncontrolled restart; the cycle continued through the normal sequence.

"DB_HMI" Status ConfigValid	Bool	TRUE	"01 PLC Health	
"DB_HMI" Status ConfigInCode	DEC	0	"01 PLC Health	
"DB_HMI" Cmd ModeRequest	DEC	0	"02 Commands	0 DB: 1 Manual, 2 Auto
"DB_HMI" Cmd Start	Bool	FALSE	"02 Commands	
"DB_HMI" Cmd Stop	Bool	FALSE	"02 Commands	
"DB_HMI" Cmd ResetFaults	Bool	FALSE	"02 Commands	
"DB_HMI" Cmd ResetCycle	Bool	FALSE	"02 Commands	
"DB_HMI" Cmd ResetSimulation	Bool	FALSE	"02 Commands	
"DB_HMI" Status System Ready	Bool	TRUE	"03 System Status	
"DB_HMI" Status System ControlEnabled	Bool	TRUE	"03 System Status	0 DB: 1 Manual, 2 Auto
"DB_HMI" Status System ActiveMode	DEC	0	"03 System Status	
"DB_HMI" Status System Offline	Bool	FALSE	"03 System Status	
"DB_HMI" Status System ManualActive	Bool	FALSE	"03 System Status	
"DB_HMI" Status System Automatic	Bool	TRUE	"03 System Status	
"DB_HMI" Status System Faulted	Bool	FALSE	"03 System Status	
"DB_HMI" Status Sequence Step	DEC	20	"04 Sequence	
"DB_HMI" Status Sequence Active	Bool	FALSE	"04 Sequence	
"DB_HMI" Status Sequence Complete	Bool	FALSE	"04 Sequence	
"DB_HMI" Status Sequence Timeout	Bool	FALSE	"04 Sequence	
"DB_HMI" Status Sequence TimeoutStep	DEC	0	"04 Sequence	
"DB_HMI" Status Sequence StepElapsedTime	Float	11.705	"04 Sequence	
"DB_HMI" Status Sequence LastCycleTimeSec	Float	95.19912	"04 Sequence	
"DB_HMI" Status Sequence CycleCounter	DEC	2	"04 Sequence	
"DB_HMI" Status Tank[1] LevelPct	Float	80.0	"05 T102 Level	
"DB_HMI" Status Tank[2] LevelPct	Float	0.0	"05 T102 Level	
"DB_HMI" Status Tank[3] LevelPct	Float	0.0	"05 T102 Level	
"DB_OP" ProcessOutValveOpenCmd[1]	Bool	FALSE	"06 Process Commands	
"DB_OP" ProcessOutValveOpenCmd[2]	Bool	FALSE	"06 Process Commands	
"DB_OP" ProcessOutValveOpenCmd[3]	Bool	FALSE	"06 Process Commands	
"DB_OP" ProcessOutPumpStartCmd[1]	Bool	FALSE	"06 Process Commands	
"DB_OP" ProcessOutPumpStartCmd[2]	Bool	FALSE	"06 Process Commands	
"DB_OP" PhysicalOutValveOpenCmd[1]	Bool	FALSE	"07 Physical Output Gate	
"DB_OP" PhysicalOutValveOpenCmd[2]	Bool	FALSE	"07 Physical Output Gate	
"DB_OP" PhysicalOutValveOpenCmd[3]	Bool	FALSE	"07 Physical Output Gate	
"DB_OP" PhysicalOutPumpStartCmd[1]	Bool	FALSE	"07 Physical Output Gate	
"DB_OP" PhysicalOutPumpStartCmd[2]	Bool	FALSE	"07 Physical Output Gate	
"DB_Alarms" Alarm ActiveWord	Hex	1640000_0000	"08 Alarm Summary	
"DB_Alarms" Alarm AnyAlarm	Bool	FALSE	"08 Alarm Summary	
"DB_Alarms" Alarm AnyTrip	Bool	FALSE	"08 Alarm Summary	
"DB_Alarms" Alarm FirstOutCode	DEC	0	"08 Alarm Summary	
"PLDQ_XV102_OpenCmd"	%Q200.0	Bool		
"PLDQ_XV102_OpenCmd"	%Q200.1	Bool		
"PLDQ_XV102_OpenCmd"	%Q200.2	Bool		
"PLDQ_XV102_StartCmd"	%Q200.3	Bool		
"DB_HMI" Status Valve[1] Fault	Bool	FALSE		
"DB_HMI" Status Valve[1] FaultCode	DEC	0		
"DB_Simulation" InjectFaultValveOpenP[1]	Bool	FALSE		
"DB_Simulation" InjectFaultValveOpenP[2]	Bool	FALSE		
"DB_Simulation" InjectFaultValveOpenP[3]	Bool	FALSE		
"DB_HMI" Status Valve[3] Fault	Bool	FALSE		
"DB_HMI" Status Valve[3] FaultCode	DEC	0		
"DB_OP" SelectedValve[3] OpenPb	Bool	FALSE		
"DB_OP" SelectedPump[2] RunPb	Bool	FALSE		
"DB_Simulation" InjectTripPumpOverhaul[2]	Bool	FALSE		
"DB_HMI" Status Pump[2] Fault	Bool	FALSE		
"DB_HMI" Status Pump[2] FaultCode	DEC	0		
"DB_OP" SelectedPump[2] OverhaulOK	Bool	TRUE		
"DB_OP" SelectedPump[2] RunPb	Bool	FALSE		
"DB_Simulation" InjectTripSafetyCircuit	Bool	FALSE		
"DB_OP" SelectedSafetyCircuitOK	Bool	TRUE		

TT-F29-A

Automatic cycle remained active at S20 with T101 at 80% and no fault/trip after the simulation-reset test action.

"DB_HMI" Status ConfigValid	Bool	TRUE	"01 PLC Health	
"DB_HMI" Status ConfigInCode	DEC	0	"01 PLC Health	
"DB_HMI" Cmd ModeRequest	DEC	0	"02 Commands	0 DB: 1 Manual, 2 Auto
"DB_HMI" Cmd Start	Bool	FALSE	"02 Commands	
"DB_HMI" Cmd Stop	Bool	FALSE	"02 Commands	
"DB_HMI" Cmd ResetFaults	Bool	FALSE	"02 Commands	
"DB_HMI" Cmd ResetCycle	Bool	FALSE	"02 Commands	
"DB_HMI" Cmd ResetSimulation	Bool	FALSE	"02 Commands	
"DB_HMI" Status System Ready	Bool	TRUE	"03 System Status	
"DB_HMI" Status System ControlEnabled	Bool	TRUE	"03 System Status	0 DB: 1 Manual, 2 Auto
"DB_HMI" Status System ActiveMode	DEC	0	"03 System Status	
"DB_HMI" Status System Offline	Bool	FALSE	"03 System Status	
"DB_HMI" Status System ManualActive	Bool	FALSE	"03 System Status	
"DB_HMI" Status System Automatic	Bool	TRUE	"03 System Status	
"DB_HMI" Status System Faulted	Bool	FALSE	"03 System Status	
"DB_HMI" Status Sequence Step	DEC	40	"04 Sequence	
"DB_HMI" Status Sequence Active	Bool	FALSE	"04 Sequence	
"DB_HMI" Status Sequence Complete	Bool	FALSE	"04 Sequence	
"DB_HMI" Status Sequence Timeout	Bool	FALSE	"04 Sequence	
"DB_HMI" Status Sequence TimeoutStep	DEC	0	"04 Sequence	
"DB_HMI" Status Sequence StepElapsedTime	Float	1.757	"04 Sequence	
"DB_HMI" Status Sequence LastCycleTimeSec	Float	95.19912	"04 Sequence	
"DB_HMI" Status Sequence CycleCounter	DEC	2	"04 Sequence	
"DB_HMI" Status Tank[1] LevelPct	Float	80.0	"05 T102 Level	
"DB_HMI" Status Tank[2] LevelPct	Float	41.5	"05 T102 Level	
"DB_HMI" Status Tank[3] LevelPct	Float	0.0	"05 T102 Level	
"DB_OP" ProcessOutValveOpenCmd[1]	Bool	FALSE	"06 Process Commands	
"DB_OP" ProcessOutValveOpenCmd[2]	Bool	TRUE	"06 Process Commands	
"DB_OP" ProcessOutValveOpenCmd[3]	Bool	FALSE	"06 Process Commands	
"DB_OP" ProcessOutPumpStartCmd[1]	Bool	FALSE	"06 Process Commands	
"DB_OP" ProcessOutPumpStartCmd[2]	Bool	FALSE	"06 Process Commands	
"DB_OP" PhysicalOutValveOpenCmd[1]	Bool	FALSE	"07 Physical Output Gate	
"DB_OP" PhysicalOutValveOpenCmd[2]	Bool	FALSE	"07 Physical Output Gate	
"DB_OP" PhysicalOutValveOpenCmd[3]	Bool	FALSE	"07 Physical Output Gate	
"DB_OP" PhysicalOutPumpStartCmd[1]	Bool	FALSE	"07 Physical Output Gate	
"DB_OP" PhysicalOutPumpStartCmd[2]	Bool	FALSE	"07 Physical Output Gate	
"DB_Alarms" Alarm ActiveWord	Hex	1640000_0000	"08 Alarm Summary	
"DB_Alarms" Alarm AnyAlarm	Bool	FALSE	"08 Alarm Summary	
"DB_Alarms" Alarm AnyTrip	Bool	FALSE	"08 Alarm Summary	
"DB_Alarms" Alarm FirstOutCode	DEC	0	"08 Alarm Summary	
"PLDQ_XV101_OpenCmd"	%Q200.0	Bool		
"PLDQ_XV102_OpenCmd"	%Q200.1	Bool		
"PLDQ_XV103_OpenCmd"	%Q200.2	Bool		
"PLDQ_XV102_StartCmd"	%Q200.3	Bool		
"DB_HMI" Status Valve[1] Fault	Bool	FALSE		
"DB_HMI" Status Valve[1] FaultCode	DEC	0		
"DB_Simulation" InjectFaultValveOpenP[1]	Bool	FALSE		
"DB_Simulation" InjectFaultValveOpenP[2]	Bool	FALSE		
"DB_Simulation" InjectFaultValveOpenP[3]	Bool	FALSE		
"DB_HMI" Status Valve[3] Fault	Bool	FALSE		
"DB_HMI" Status Valve[3] FaultCode	DEC	0		
"DB_OP" SelectedValve[3] OpenPb	Bool	FALSE		
"DB_OP" SelectedPump[2] RunPb	Bool	FALSE		
"DB_Simulation" InjectTripPumpOverhaul[2]	Bool	FALSE		
"DB_HMI" Status Pump[2] Fault	Bool	FALSE		
"DB_HMI" Status Pump[2] FaultCode	DEC	0		
"DB_OP" SelectedPump[2] OverhaulOK	Bool	TRUE		
"DB_OP" SelectedPump[2] RunPb	Bool	FALSE		
"DB_Simulation" InjectTripSafetyCircuit	Bool	FALSE		
"DB_OP" SelectedSafetyCircuitOK	Bool	TRUE		

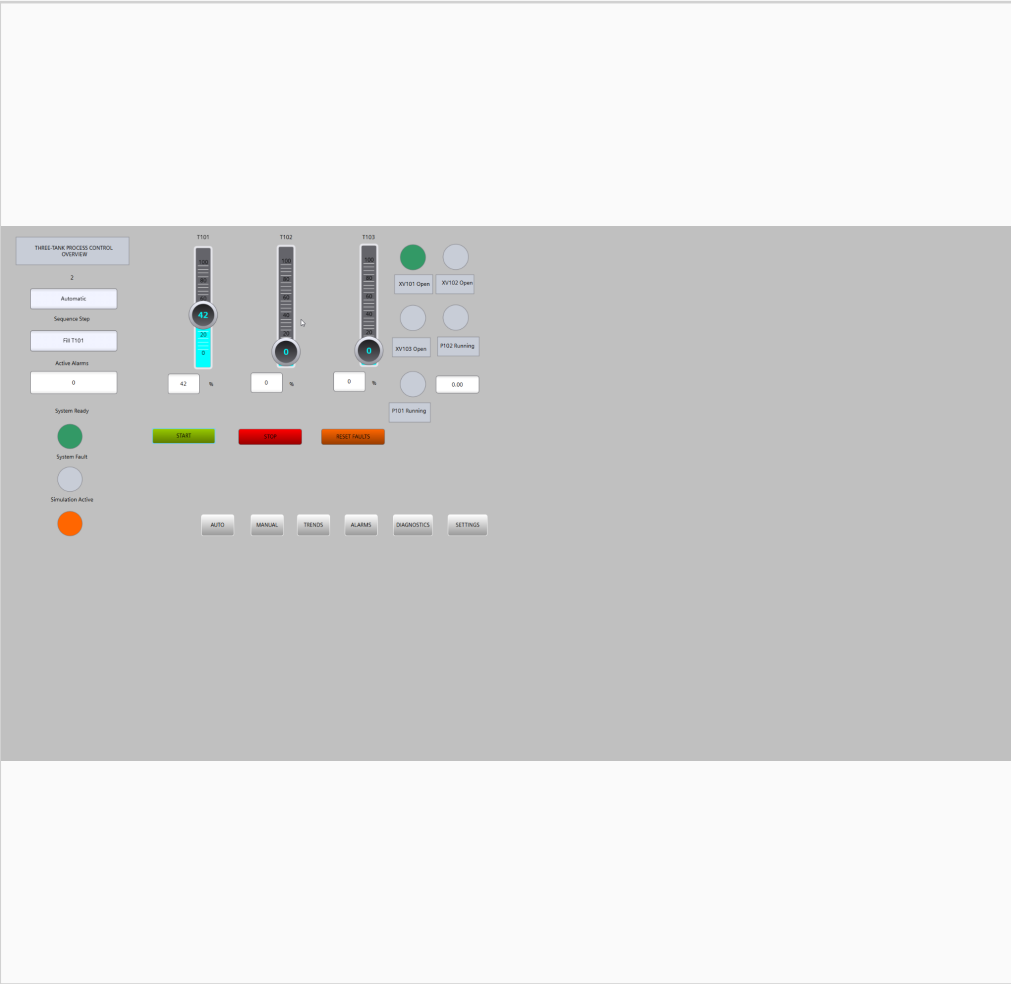
TT-F29-B

The same cycle progressed normally to S40 with T102 rising and the XV102/P101 process path active.

- Check

Start a clean cycle after cycle reset
- Observed

ResetCycle returned the model to idle; a deliberate Auto + Start then entered precheck and S20 with T101 rising.



TT-F30 HMI
Fresh cycle in S20 after cycle reset and deliberate restart.



TT-F30 watch
PLC state for the fresh S20 cycle.